Cisco 200-901

DevNet Associate (DEVASC) · 631 questions (with answers)

QUESTION 1

Issued to kamjunhao@gmail.com · October 17, 2026

Which two concepts describe test-driven development? (Choose two.)

A User acceptance testers develop the test requirements.

B It enables code refactoring. ✓ CORRECT

C Tests are created when code is ready for release.

D Implementation is driven by incremental testing of release candidates.

E Write a test before writing code. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is B, E.

Test-driven development (TDD) is a software development methodology that involves writing a test before writing the actual code. This ensures that the code is tested incrementally and helps in refining and improving it through continuous refactoring. By following these steps, TDD facilitates the creation of clean, efficient, and bug-free code. Therefore, enabling code refactoring and writing a test before writing code are two key concepts that describe TDD.

Which task is performed because the test-driven development approach is being used?

A creating test scenarios based on continuous development

B writing code without committing any coding violations

C refactoring code that is covered by existing tests ✓ CORRECT

D testing existing software before developing new code

SUGGESTED ANSWER

The suggested answer is C.

In test-driven development (TDD), the process involves writing tests before writing the actual code. Once the test is written and fails (because the functionality is not yet implemented), the developer writes the minimum amount of code required for the test to pass. After the test passes, the code is then refactored to ensure it is clean and efficient while still passing the tests. The task of refactoring code that is covered by existing tests aligns with the principles of TDD, as it involves refining the code after it has already been tested to ensure it meets all requirements and maintains readability and structure without changing its behavior.

What is the **FIRST** development task in test-driven development?

A Write code that implements a desired function.

B Write a failing test case for a desired function. ✓ CORRECT

C Reverse engineer the code for a desired function.

D Write a passing test case for existing code.

SUGGESTED ANSWER

The suggested answer is B.

In test-driven development (TDD), the first development task is to write a failing test case for the desired function. This initial test case is intended to define the function's expected behavior before any actual implementation code is written, ensuring that the code development follows the requirements specified by the test. Only after writing this failing test and seeing it fail does the developer proceed to write the minimum amount of code necessary to make the test pass.

examice.com

Issued to kamjunhao@gmail.com · October 17, 2026

In test-driven development, what are two of the green bar patterns? (Choose two.)

A another test

B break

C triangulate ✓ CORRECT

D starter test

E fake it ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is C, E.

In test-driven development (TDD), two recognized green bar patterns are 'triangulate' and 'fake it'. 'Triangulate' refers to the method of using multiple tests to drive the design of the code, while 'fake it' is about implementing simple code that makes the test pass in the simplest way possible. These patterns help ensure that the code evolves incrementally and maintains a clear focus on passing tests while gradually improving the implementation.

Issued to kamjunhao@gmail.com · October 17, 2026

In the test-driven development model, what is changed after a test fails?

A schedule

B project requirements

C code ✓ CORRECT

D test

SUGGESTED ANSWER

The suggested answer is C.

In the test-driven development model, after a test fails, the code is changed. This is because the primary purpose of test-driven development is to write the minimum amount of code necessary to pass the failing test. The process involves writing a test that initially fails, then writing the code to make the test pass, followed by refactoring the code if necessary. Therefore, the correct answer is that the code is changed.

Issued to kamjunhao@gmail.com · October 17, 2026

What is a benefit of test-driven development?

A strict adherence to product requirements

B faster releases that have minimal features

C early customer involvement

D increased code quality ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

A key benefit of test-driven development is increased code quality. By writing tests before the actual production code, developers are able to identify and fix defects early in the development process, leading to cleaner, more reliable, and maintainable code. TDD also encourages better design and helps ensure that the code meets its requirements by verifying functionality through automated testing.

Issued to kamjunhao@gmail.com · October 17, 2026

Which two statements describe the advantages of using a version control system? (Choose two.)

A. It allows for branching and merging so that different tasks are worked on in isolation before they are merged into a feature or master branch. ✓ CORRECT

B. It provides tooling to automate application builds and infrastructure provisioning.

C. It allows multiple engineers to work against the same code and configuration files and manage differences and conflicts. ✓ CORRECT

D. It provides a system to track User Stories and allocate to backlogs.

E. It allows developers to write effective unit tests.

SUGGESTED ANSWER

The suggested answer is A, C.

Using a version control system provides several advantages. Firstly, it allows for branching and merging, which means different tasks can be worked on independently before being merged into a main branch. This helps manage contributions from multiple developers and maintain code integrity. Secondly, it enables multiple engineers to collaborate on the same codebase simultaneously, effectively managing differences and conflicts through version control mechanisms.

Personal copy · kamjunhao@gmail.com

Issued to kamjunhao@gmail.com · October 17, 2026

What are two advantages of version control software? (Choose two.)

A It supports tracking and comparison of changes in binary format files.

B It allows new team members to access the current code and history. ✓ CORRECT

C It supports comparisons between revisions of source code files. ✓ CORRECT

D It provides wiki collaboration software for documentation.

E It allows old versions of packaged applications to be hosted on the Internet.

SUGGESTED ANSWER

The suggested answer is B, C.

Version control software allows new team members to access the current code and the history of all changes made to it. This capability is crucial for collaborative development and onboarding new developers. Additionally, it supports comparisons between revisions of source code files, which is essential for tracking changes, debugging, and understanding the evolution of the codebase over time.

Issued to kamjunhao@gmail.com · October 17, 2026

Before which process is `code` review performed when `version` control is used?

A checkout of code

B merge of code ✓ CORRECT

C committing code

D branching code

SUGGESTED ANSWER

The suggested answer is B.

Code review is performed before merging code. During the version control process, changes are typically made in separate branches. Once the changes are ready, a merge request or pull request is created, which initiates the code review process. The code must be reviewed and approved before it can be merged into the main branch or another integration branch. This ensures that the code being merged meets the project's standards and does not introduce defects.

Issued to kamjunhao@gmail.com · October 17, 2026

What is an advantage of a version control system?

A facilitates resolving conflicts when merging code ✓ CORRECT

B ensures that unit tests are written

C prevents over-writing code or configuration files

D forces the practice of trunk-based development

SUGGESTED ANSWER

The suggested answer is A.

An advantage of a version control system is that it facilitates resolving conflicts when merging code. Version control systems allow multiple developers to work on the same codebase simultaneously by managing changes and integrating those changes cohesively. This enables developers to detect conflicts early and provides tools to resolve them, ensuring smooth collaboration. By tracking changes, version control systems also allow reversing modifications and understanding the history of a project, which are critical aspects for maintaining and developing software effectively.

Refer to the exhibit.

```
json_string = """
{
        "researcher": {
            "name": "Ford Perfect",
            "species": "Betelgeusian",
            "relatives": [
                {
                    "name": "Zaphod Beeblebrox",
                    "species": "Betelgeusian"
                }
            ]
        }
}
"""

my_json = json.loads(json_string)
```

Which Python data structure does my_json contain?

A map

B list

C json

D dict ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

The given JSON string is a nested structure with dictionaries and lists. When using the json.loads function in Python with a JSON string, it converts the JSON string into a corresponding Python data structure. In this case, the outer structure of the JSON is an object, which converts into a Python dictionary. Thus, the my_json variable will contain a dictionary after the json.loads operation.

QUESTION 12

Issued to kamjunhao@gmail.com · October 17, 2026

When a Cisco IOS XE networking device is configured using RESTCONF, what is the default data-encoding method?

A JSON

B YAML

C XML ✓ CORRECT

D x-form-encoding

SUGGESTED ANSWER

The suggested answer is C.

When a Cisco IOS XE networking device is configured using RESTCONF, the default data-encoding method is XML. RESTCONF supports both JSON and XML, but in the absence of an 'Accept' header specifying the format, the system defaults to XML. This can be observed by performing a cURL request without specifying the 'Accept' header, which returns data encoded in XML format.

Refer to the exhibit.

```
<books>
 <science>
  <biology>10.00</biology>
  <geology>9.00</geology>
 </science>
 <math>
  <calculus>20.00</calculus>
 </math>
</books>
```

Which JSON is equivalent to the XML-encoded data?

- A.

```
[ {
  "books": {
    "science": {
      "biology": "10.00",
      "geology": "9.00",
    },
    "math": {
      "calculus": "20.00",
    }
  }
} ]
```

- B.

```
{
   "books": {
      "science": {
         "biology": "10.00",
         "geology": "9.00",
      },
      "math": {
         "calculus": "20.00",
      }
   }
}
```

- C.

```
{
    "books": [
      "science": {
         "biology": "10.00",
         "geology": "9.00",
      },
      "math": {
         "calculus": "20.00",
      }
   ]
}
```

- D.

```
{
    "books": [
      "science", {
         "biology": "10.00",
         "geology": "9.00",
      },
      "math", {
         "calculus": "20.00",
      }
   ]
}
```

SUGGESTED ANSWER

The XML data provided describes a collection of books with two categories: 'science' and 'math'. Within each category, there are individual books with their respective prices. The correct JSON representation should reflect this hierarchical structure without introducing any unnecessary arrays or altering the XML structure. Therefore, the correct JSON representation of the given XML should be structured as follows: { "books": { "science": { "biology": "10.00", "geology": "9.00" }, "math": { "calculus": "20.00" } } }. This matches option B, which accurately represents the structure and data defined by the original XML.

QUESTION 14

FILL BLANK -

Fill in the blanks to complete the Python script to request a service ticket using the APIC-EM REST API for the user `devnetuser`. import requests import json controller = 'devnetapi.cisco.com/sandbox/apic_em' url = `https://` + controller + `api/va/ticket` payload = {'username': '_____', 'password': '370940885'} header = {'Content-type': 'application.json'} response = _____.post(url, data=json.dumps(payload), \ headers= _____, verify=False) r_json = response.json() print(r_json) ticket = r_json[`response`][`serviceTicket`] print(ticket)

SUGGESTED ANSWER

To complete the Python script for requesting a service ticket using the APIC-EM REST API for the user 'devnetuser', you need to fill in the blanks with appropriate values. The 'username' in the payload should be 'devnetuser'. The `requests` library should be used for making the HTTP request. The 'header' should specify the content type as 'application/json'. Here is the completed script: import requests import json controller = 'devnetapi.cisco.com/sandbox/apic_em' url = 'https://' + controller + 'api/va/ticket' payload = {'username': 'devnetuser', 'password': '370940885'} header = {'Content-type': 'application/json'} response = requests.post(url, data=json.dumps(payload), headers=header, verify=False) r_json = response.json() print(r_json) ticket = r_json['response'] ['serviceTicket'] print(ticket)

Issued to kamjunhao@gmail.com · October 17, 2026

Which two statements about JSON and XML are

true? (Choose two.)

A The syntax of JSON contains tags, elements, and attributes.

B XML objects are collections of key-value pairs.

C JSON objects are collections of key-value pairs. ✓ CORRECT

D JSON arrays are an unordered set of key-value pairs.

E The syntax of XML contains tags, elements, and attributes. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is C, E.

JSON objects are collections of key-value pairs, which make them ideal for representing structured data. XML, on the other hand, uses a syntax that includes tags, elements, and attributes to define data structure. These two features highlight fundamental differences in how JSON and XML store and represent data.

Which statement describes the benefit of using functions in programming?

A Functions ensure that a developer understands the inner logic contained before using them as part of a script or application.

B Functions create the implementation of secret and encrypted algorithms.

C Functions allow problems to be split into simpler, smaller groups, and reduce code repetition, which makes the code easier to read. ✓ CORRECT

D Functions store mutable values within a script or application.

SUGGESTED ANSWER

The suggested answer is C.

Functions allow problems to be split into simpler, smaller groups, and reduce code repetition, which makes the code easier to read.

examice.com

Refer to the exhibit.

```
$ diff -u5 fish.py cat.py
--- fish.py     2020-01-02 09:41:02.840000000 +0100
+++ cat.py      2020-01-02 09:41:06.8859999800 +0100
@@ -160,11 +160,12 @@

     @single_request_timeout.setter
     def single_request_timeout(self, value):
         """The timeout (seconds) for a single HTTP REST API request."""
         check_type(value, int, optional=True)
-        assert value is None or value > 0
+        if value is not None and value <= 0:
+            raise ValueError("single_request_timeout must be positive integer")
         self._single_request_timeout = value


     @property
     def wait_on_rate_limit(self)
         """Automatic rate-limit handling.
```

The output of a unified diff when comparing two versions of a Python script is shown. Which two `single_request_timeout()` functions are defined in fish.py and cat.py? (Choose two.)

- A.

file: cat.py

```
160
161     @single_request_timeout.setter
162     def single_request_timeout(self, value):
163         """The timeout (seconds) for a single HTTP REST API request."""
164         check_type(value, int, optional=True)
165         assert value is None or value > 0
166         if value is not None and value <= 0:
167             raise ValueError("single_request_timeout must be positive integer")
168         self._single_request_timeout = value
169
```

- B.

file: fish.py

```
160
161     @single_request_timeout.setter
162     def single_request_timeout(self, value):
163         """The timeout (seconds) for a single HTTP REST API request."""
164         check_type(value, int, optional=True)
165         assert value is None or value > 0
166         self._single_request_timeout = value
167
```

- C.

file: cat.py

```
160
161     @single_request_timeout.setter
162     def single_request_timeout(self, value):
163         """The timeout (seconds) for a single HTTP REST API request."""
164         check_type(value, int, optional=True)
165         if value is not None and value <= 0:
166             raise ValueError("single_request_timeout must be positive integer")
167         self._single_request_timeout = value
168
169
```

- D.

file: fish.py

```
160
161     @single_request_timeout.setter
162     def single_request_timeout(self, value):
163         """The timeout (seconds) for a single HTTP REST API request."""
164         check_type(value, int, optional=True)
165         self._single_request_timeout = value
```

- E.

file: cat.py

```
172
173     @single_request_timeout.setter
174     def single_request_timeout(self, value):
175         """The timeout (seconds) for a single HTTP REST API request."""
176         check_type(value, int, optional=True)
177         if value is not None and value <= 0:
178             raise ValueError("single_request_timeout must be positive integer")
179         self._single_request_timeout = value
180
```

SUGGESTED ANSWER

The exhibit shows a unified diff output comparing two versions of a Python script, `fish.py` and `cat.py`. From the diff output, we can observe the changes made to the `single_request_timeout` function. The function definition in `fish.py` defines an assertion `assert value is None or value > 0`, whereas in `cat.py`, this assertion is replaced with an `if` statement that raises a `ValueError` if the condition `value is not None and value <= 0` is met. Therefore, the correct `single_request_timeout()` function definitions in `fish.py` and `cat.py` are as follows: for `fish.py`, the function definition matches option B, and for `cat.py`, the function definition matches option C.

QUESTION 18

Issued to kamjunhao@gmail.com · October 17, 2026

What is a benefit of organizing code into modules?

A reduces the length of code

B enables code to be multifunctional

C enables the reuse of code ✓ CORRECT

D improves overall performance

SUGGESTED ANSWER

The suggested answer is C.

Organizing code into modules primarily benefits software development by enabling the reuse of code. Modules encapsulate specific functionality in isolated units, which makes it possible to use the same code segment in multiple parts of a project without duplication. This leads to increased maintainability and efficiency in code management. Additionally, reusable modules help in simplifying debugging, testing, and updating processes.

Issued to kamjunhao@gmail.com · October 17, 2026

What is a benefit of organizing code into modules?

A enables the code to be broken down into layers

B improves collaboration of the development team

C makes it easier to deal with large and complex systems ✓ CORRECT

D enables the inclusion of more programming languages in the code

SUGGESTED ANSWER

The suggested answer is C.

One of the primary benefits of organizing code into modules is that it significantly simplifies the management of large and complex systems. By breaking down code into smaller, self-contained pieces, modules make the codebase more manageable, easier to understand, and simpler to maintain. This modular structure also supports code reusability and separation of concerns, which enhances the overall development process.

Issued to kamjunhao@gmail.com · October 17, 2026

What is the Git command to delete a local branch named `experiment` without a warning?

A git branch λ€"rm experiment

B git branch λ€"n experiment

C git branch λ€"f experiment

D git branch λ€"D experiment ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

The Git command to delete a local branch named 'experiment' without any warning is 'git branch -D experiment'. The '-D' flag stands for delete force, which ensures the branch is removed without any warnings or confirmations. While the '-d' option is also used to delete a branch, it would prompt a warning if the branch has not been merged. Therefore, '-D' is the appropriate option for deleting without a warning.

Issued to kamjunhao@gmail.com · October 17, 2026

What is the outcome of executing this command?

`git clone ssh:/john@example.com/path/to/my-project.git`

A creates a local copy of a repository called ɹ€my-projectɹ€ ✓ CORRECT

B initiates a new Git repository called ɹ€my-projectɹ€

C creates a copy of a branch called ɹ€my-projectɹ€

D creates a new branch called ɹ€my-projectɹ€

SUGGESTED ANSWER

The suggested answer is A.

The 'git clone' command is used to create a local copy of an existing repository. It takes a repository URL as an argument and copies the entire repository, including its history and branches, to the local machine. In this case, executing 'git clone ssh:///path/to/my-project.git' will create a local copy of the repository located at the specified URL and name the local directory 'my-project'.

Issued to kamjunhao@gmail.com · October 17, 2026

Which action does the Git command git merge allow the developer to perform?

A Combine multiple sequences of commits into one unified history ✓ CORRECT

B Push changes to the remote repository

C Create, list, rename, and delete branches

D Switch between branches

SUGGESTED ANSWER

The suggested answer is A.

The git merge command is used to combine multiple sequences of commits into one unified history. This action is commonly used when integrating changes from different branches, ensuring that all work is brought together into a single branch. It is an essential part of version control to maintain a cohesive project history.

How does a developer create and switch to a new branch called `my-bug-fix` to develop a product fix?

A git checkout -b my-bug-fix ✓ CORRECT

B git branch -b my-bug-fix

C git branch my-bug-fix

D git checkout my-bug-fix

SUGGESTED ANSWER

The suggested answer is A.

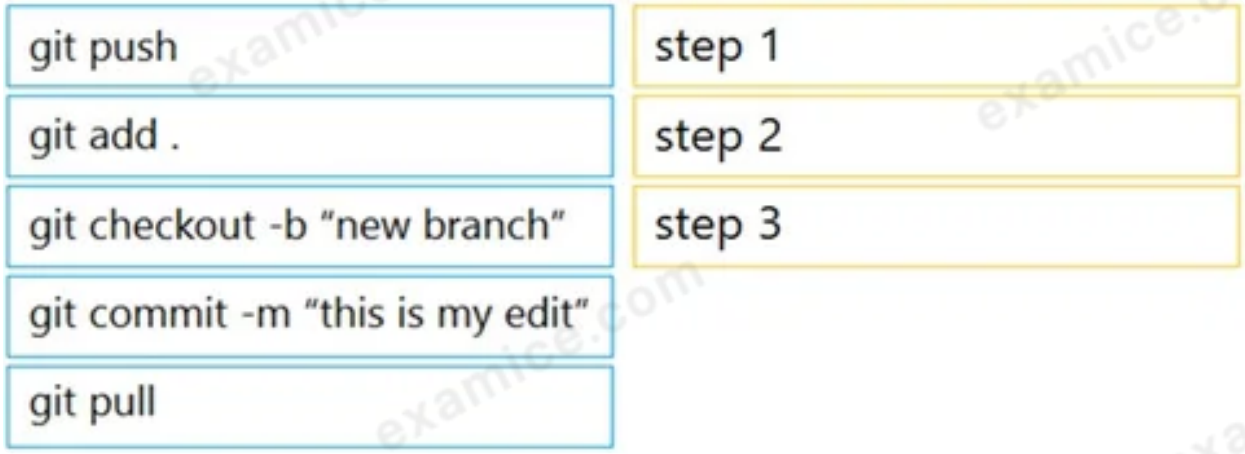
To create and switch to a new branch called 'my-bug-fix,' the correct command is 'git checkout -b my-bug-fix'. This command simultaneously creates the new branch and switches to it, making it a convenient choice for developers looking to start work on a new feature or fix.

examice.com

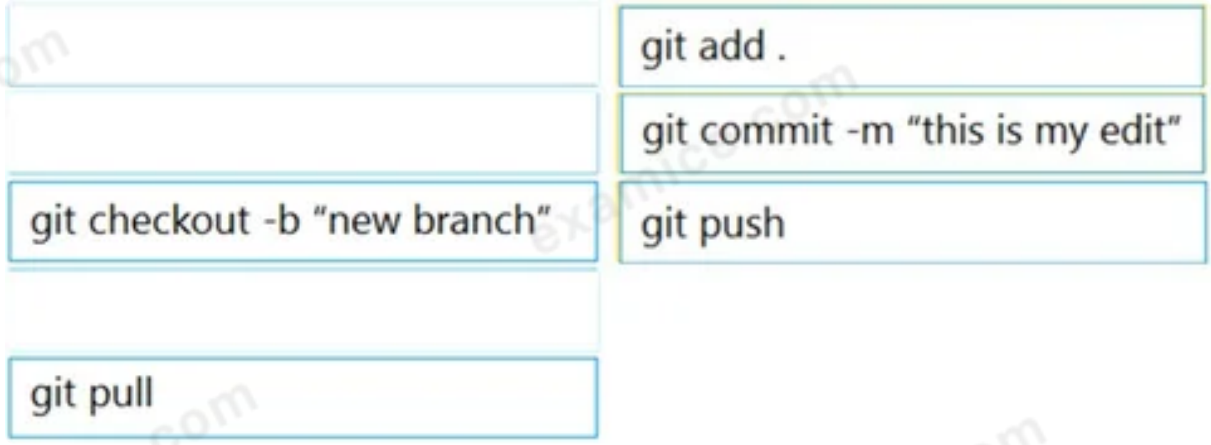
DRAG DROP

Drag and drop the Git commands from the left onto the right that add modified local files to a remote repository. Not all options are used.

Select and Place:



SUGGESTED ANSWER



Issued to kamjunhao@gmail.com · October 17, 2026

A developer needs to prepare the file README.md in the working tree for the next commit operation using Git. Which command needs to be used to accomplish this?

A git -a README.md

B git add README.md ✓ CORRECT

C git add README.md staging

D git commit README.md

SUGGESTED ANSWER

The suggested answer is B.

In order to prepare the file README.md in the working tree for the next commit, the appropriate Git command is 'git add README.md'. This command stages the file, making it ready to be committed. Without running this command, changes to README.md will not be included in the next commit.

Issued to kamjunhao@gmail.com · October 17, 2026

A developer is reviewing a code that was written by a colleague. It runs fine, but there are many lines of code to do a seemingly simple task repeatedly. Which action organizes the code?

A Refactor the code by removing any unnecessary tests.

B Reverse engineer and rewrite the code logic.

C Using functions, rewrite any pieces of code that are repeated. ✓ CORRECT

D Modify the code to use loops.

SUGGESTED ANSWER

The suggested answer is C.

When a piece of code is repeated multiple times, the appropriate action to organize and streamline the code is to encapsulate the repeated code into a function. This not only reduces duplication but also makes the code more modular and easier to maintain. Using functions allows for the repeated logic to be written once and then called whenever needed, resulting in cleaner and more organized code.

Which principle is a value from the manifesto for Agile software development?

A processes and tools over teams and interactions

B detailed documentation over working software

C adhering to a plan over responding to requirements

D customer collaboration over contract negotiation ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

The Agile Manifesto promotes principles that facilitate agile development by emphasizing values such as customer collaboration over contract negotiation. This principle underscores the importance of working closely with the customer to fulfill their needs and adapt to changing requirements, rather than rigidly adhering to predefined contracts.

examice.com

Issued to kamjunhao@gmail.com · October 17, 2026

Which advantage does the agile process offer compared to waterfall software development?

A to add or update features with incremental delivery ✓ CORRECT

B to view the full scope of end-to-end work

C to have each phase end before the next begins

D to fix any issues at the end of the development cycle

SUGGESTED ANSWER

The suggested answer is A.

The agile process allows for the addition or updating of features with incremental delivery. This approach involves working in smaller time increments called 'sprints,' where each sprint encompasses the full process of analysis, design, coding, and testing for a small aspect of the application. This enables teams to deliver functional pieces of the software incrementally and make modifications based on feedback, improving flexibility and adaptability compared to the waterfall process.

Issued to kamjunhao@gmail.com · October 17, 2026

How do XML and JSON compare regarding functionality?

A XML provides more support for mapping data structures into host languages than JSON.

B XML provides more human readability than JSON.

C JSON provides less support for data types than XML.

D JSON natively supports arrays and XML does not natively support arrays. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

JSON natively supports arrays through the use of square brackets, making it straightforward to represent list-like structures. XML does not have a native array construct; instead, it uses repeating elements with the same name to represent arrays, which requires more complex parsing.

What are two principles of an infrastructure as code environment? (Choose two.)

A Components are coupled, and definitions must be deployed for the environment to function.

B Redeployments cause varying environment definitions.

C Environments must be provisioned consistently using the same inputs. ✓ CORRECT

D Service overlap is encouraged to cater for unique environment needs.

E Complete complex systems must be able to be built from reusable infrastructure definitions. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is C, E.

Two principles of an infrastructure as code environment are that environments must be provisioned consistently using the same inputs and that complete complex systems must be able to be built from reusable infrastructure definitions. This ensures repeatability, consistency, and scalability in managing infrastructure, which are key in an infrastructure as code setup.

Issued to kamjunhao@gmail.com · October 17, 2026

Which two encoding formats do YANG interfaces support? (Choose two.)

A XML ✓ CORRECT

B JSON ✓ CORRECT

C XHTML

D BER

E plain text

SUGGESTED ANSWER

The suggested answer is A, B.

YANG data models are typically used in network management protocols like NETCONF and RESTCONF. These protocols support data encoding formats such as XML and JSON to represent the YANG data models. Thus, XML and JSON are the two encoding formats that YANG interfaces support.

Refer to the exhibit.

```
{
  "items": [{
     "kind": "object#NetworkObj",
     "selfLink": "https://10.201.230.5/api/objects/networkobjects/db01",
     "name": "db01",
     "host": {
        "kind": "IPv4address",
        "value": "172.16.0.11"
     },
     "objectId": "db01"
  }]
}
```

The JSON data in the exhibit has been parsed and stored into a variable `data`. What returns the value `172.16.0.11`?

A data['items']['host']['value']

B data['items'][1]['host']['value']

C data['items'][0]['host']['value'] ✓ CORRECT

D data['items']['host'][1]

SUGGESTED ANSWER

The suggested answer is C.

To access the value `172.16.0.11`, we need to navigate through the nested structure of the parsed JSON data stored in the variable `data`. The `items` key contains a list, and we need to access the first element of this list, which is at index 0. Within this element, we then need to access the `host` key, and finally the `value` key. Therefore, the correct expression is data['items'][0]['host']['value'].

Refer to the exhibit.

```
def get_result()
        url = "https://sandboxdnac.cisco.com/dna/system/api/v1/auth/token"
        resp = requests.post(url, auth=HTTPBasicAuth(DNAC_USER, DNAC_PASSWORD))
        result = resp.json()['Token']
        return result
```

What does the Python function do?

A It returns HTTP Basic Authentication.

B It returns DNAC user and password.

C It reads a token from a local JSON file and posts the token to the DNAC URL.

D It returns an authorization token. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

The Python function defined in the exhibit makes an HTTP POST request to a specified URL using HTTP Basic Authentication. It then retrieves and returns an authorization token from the server's JSON response. This is clear from the line `result = resp.json()['Token']`, which indicates that the token is extracted from the JSON response of the POST request. Therefore, the correct answer is that the function returns an authorization token.

Package updates from a local server fail to download. However, the same updates work when a much slower external repository is used. Why are local updates failing?

A The server is running out of disk space.

B The Internet connection is too slow.

C The Internet is down at the moment, which causes the local server to not be able to respond.

D The update utility is trying to use a proxy to access the internal resource. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

The issue described likely stems from the update utility trying to use a proxy to access the internal resource. If a proxy configuration is set incorrectly or if it is attempting to route through an external network to communicate with a local server, it could cause the local updates to fail while still being able to connect to the slower external repository.

examice.com

Issued to kamjunhao@gmail.com · October 17, 2026

What is a functionality of the Waterfall method as compared to the Agile method for software development?

A Waterfall increases agility to implement faster while Agile promotes reliability.

B A phase begins after the previous phase has ended in Waterfall while Agile phases run in parallel. ✓ CORRECT

C Customers get feedback during the process in Waterfall while they can see the result at the end in Agile.

D Requirements can be updated in Waterfall while in Agile it should be gathered in the beginning.

SUGGESTED ANSWER

The suggested answer is B.

In the Waterfall method, each phase of the software development process must be completed before the next phase begins. This rigid sequential approach contrasts with Agile, where phases can overlap and development cycles are iterative, allowing for more flexibility and adaptability throughout the process.

Issued to kamjunhao@gmail.com · October 17, 2026

What is a benefit of using functions in the code for the development process?

A better user experience in the end product

B improves code performance

C easier to compile the code

D faster code development ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

Using functions in code helps streamline the development process by allowing developers to write reusable blocks of code. This enables faster code development, as developers can avoid writing repetitive code and focus on implementing new features or fixing issues more efficiently.

Issued to kamjunhao@gmail.com · October 17, 2026

What is the difference between YAML and JSON data structure?

A YAML uses spaces; JSON uses parentheses

B YAML uses indentation; JSON uses brackets and braces ✓ CORRECT

C YAML uses brackets and braces; JSON uses indentation

D YAML uses parentheses; JSON uses spaces

SUGGESTED ANSWER

The suggested answer is B.

YAML uses indentation to indicate the structure of the data, with levels of nesting represented by the indentation level. JSON, on the other hand, uses brackets and braces to define arrays and objects, respectively. This fundamental difference in how the structure is represented is a key distinction between the two formats.

A developer is working on a feature for a new application. The changes in the existing branch named **'feat00304'** must be integrated into a single commit with the current working primary branch named **'prodapp411926287'**. Which git command must be used?

A git rebase --merge feat00304

B git merge --squash feat00304 ✓ CORRECT

C git push --rebase feat00304

D git checkout --squash feat00304

SUGGESTED ANSWER

The suggested answer is B.

To integrate changes from an existing branch into a single commit with the current working primary branch, the appropriate git command is 'git merge --squash feat00304'. The 'git merge --squash' command allows you to combine all the changes from the specified branch into a single commit, which is exactly the requirement in this scenario.

examice.com

Issued to kamjunhao@gmail.com · October 17, 2026

A file that already exists in a local repository is updated. Which command **MUST** be executed to ensure that the changes in the file are included in the next Git commit?

A git update

B git merge

C git add ✓ CORRECT

D git rebase

SUGGESTED ANSWER

The suggested answer is C.

To ensure that changes in an updated file are included in the next Git commit, the correct command to use is 'git add'. This command stages the changes in the file so that they are ready to be committed. Other commands like 'git update', 'git merge', and 'git rebase' do not perform this function. 'git update' is not a recognized Git command, 'git merge' is used to integrate changes from different branches, and 'git rebase' is used to reapply commits on top of another base tip.

Issued to kamjunhao@gmail.com · October 17, 2026

Into which type of Python data structure should common data formats be parsed?

A sequence

B set

C dictionary ✓ CORRECT

D list

SUGGESTED ANSWER

The suggested answer is C.

When parsing common data formats, such as JSON and XML, into Python data structures, the most appropriate structure is a dictionary. Dictionaries in Python can represent nested data with key-value pairs, which aligns with how data is often structured in these formats. A dictionary allows for efficient data retrieval based on keys, making it ideal for this purpose.

Issued to kamjunhao@gmail.com · October 17, 2026

Several teams at a company are developing a new CRM solution to track customer interactions with a goal of improving customer satisfaction and driving higher revenue. The proposed solution contains these components:

- MySQL database that stores data about customers
- HTML5 and JavaScript UI that runs on Apache
- REST API written in Python

What are two advantages of applying the MVC design pattern to the development of the solution? (Choose two.)

A to enable multiple views of the same data to be presented to different groups of users ✓ CORRECT

B to provide separation between the view and the model by ensuring that all logic is separated out into the controller ✓ CORRECT

C to ensure data consistency, which requires that changes to the view are also made to the model

D to ensure that only one instance of the data model can be created

E to provide only a single view of the data to ensure consistency

SUGGESTED ANSWER

The suggested answer is A, B.

Applying the MVC design pattern to the CRM solution offers several advantages. First, it enables multiple views of the same data to be presented to different groups of users. This flexibility allows different user interfaces to be developed and maintained independently, catering to various user needs without affecting the underlying data model. Second, it provides separation between the view and the model by ensuring that all logic is separated out into the controller. This separation of concerns simplifies application development and maintenance by allowing developers to work on the UI, business logic, and data management independently.

```
module ietf-interface {
  namespace "urn:ietf:params:xml:ns:yang:ietf-interfaces";
  prefix if;
  import ietf-yang-types {
    prefix yang;
  }
  container interfaces-state {
    list interface {
      key "name";
      leaf name {
        type string;
      }
      leaf admin-status {
        type enumeration {
          enum up {
            value 1;
          }
          enum down {
            value 2;
          }
          enum testing {
            value 3;
          }
        }
      }
      leaf if-index {
        type int32 {
          range "1..2147483647"
        }
      }
      container statistics {
        leaf in-octets {
          type yang:counter64;
        }
        leaf in-unicast-pkts {
          type yang:counter64;
        }
      }
    }
  }
}
```

Refer to the exhibit. Which XML snippet has interface information that conforms to the YANG model?

- A.

```
<interfaces-state
xmlns="urn:ietf:params:xml:ns:yang:ietf-interfaces"
xmlns:if="urn:ietf:params:xml:ns:yang:ietf-interfaces">
    <interface name="GigabitEthernet1">
        <admin-status>1</admin-status>
        <if-index>1</if-index>
        <statistics>
            <in-octets>408164820</in-octets>
            <in-unicast-pkts>728061</in-unicast-pkts>
        </statistics>
    </interface>
</interfaces-state>
```

- B.

```
<interfaces-state
xmlns="urn:ietf:params:xml:ns:yang:ietf-interfaces"
xmlns:if="urn:ietf:params:xml:ns:yang:ietf-interfaces">
    <interface>
        <name>GigabitEthernet1</name>
        <admin-status>up</admin-status>
        <if-index>1</if-index>
        <in-octets>408164820</in-octets>
        <in-unicast-pkts>728061</in-unicast-pkts>
    </interface>
</interfaces-state>
```

- C.

```
<interfaces-state
xmlns="urn:ietf:params:xml:ns:yang:ietf-interfaces"
xmlns:if="urn:ietf:params:xml:ns:yang:ietf-interfaces">
    <interface>
        <name>GigabitEthernet1</name>
        <admin-status>up</admin-status>
        <if-index>1</if-index>
        <statistics>
            <in-octets>408164820</in-octets>
            <in-unicast-pkts>728061</in-unicast-pkts>
        </statistics>
    </interface>
</interfaces-state>
```

- D.

```
<interfaces-state
xmlns="urn:ietf:params:xml:ns:yang:ietf-interfaces"
xmlns:if="urn:ietf:params:xml:ns:yang:ietf-interfaces">
    <interface>
        <name>GigabitEthernet1</name>
        <admin-status>1</admin-status>>
        <if-index>1</if-index>
        <statistics>
            <in-octets>408164820</in-octets>
            <in-unicast-pkts>728061</in-unicast-pkts>
        </statistics>
    </interface>
</interfaces-state>
```

SUGGESTED ANSWER

The correct answer is C. The YANG model defines an

Personal copy · kamjunhao@gmail.com

Issued to kamjunhao@gmail.com · October 17, 2026

What is a benefit of version control?

A prevents two users from working on the same file

B keeps track of all changes to the files ✓ CORRECT

C prevents the sharing of files

D keeps the list of data types used in the files

SUGGESTED ANSWER

The suggested answer is B.

A benefit of version control is that it keeps track of all changes to the files. Version control systems allow multiple users to collaborate on a project by keeping a history of changes, enabling users to revert to previous versions if needed, and providing tools to resolve conflicts when different changes are made to the same file.

```
git clone git://git.kernel.org/.../git.git my.git
cd my.git
git branch -d -r origin/todo origin/html origin/man    (1)
git branch -D test                                     (2)
```

Refer to the exhibit. What does the command marked (2) do when it is run?

A It duplicates the ɹ€testɹ€ branch.

B It deletes the ɹ€testɹ€ branch only if a new branch is created.

C It deletes the ɹ€testɹ€ branch. ✓ CORRECT

D It does not delete the branch until it is merged.

SUGGESTED ANSWER

The suggested answer is C.

The command 'git branch -D test' forcefully deletes the branch named 'test'. The '-D' flag is used to delete the branch regardless of its merge status, unlike '-d' which deletes the branch only if it has been fully merged. Therefore, the correct answer is that it deletes the 'test' branch.

Issued to kamjunhao@gmail.com · October 17, 2026

What is a comparison of YAML and JSON?

A YAML has a more consistent approach to representing data compared to JSON.

B JSON does not support comments and YAML does. ✓ CORRECT

C YAML is a more verbose data structure compared to JSON.

D JSON has more common usage in configuration management tools compared to YAML.

SUGGESTED ANSWER

The suggested answer is B.

JSON does not support comments, while YAML does. This is a fundamental difference between the two formats. JSON is designed to be a data-only format, whereas YAML is intended to be readable by humans and thus includes the possibility of comments to annotate the data.

Issued to kamjunhao@gmail.com · October 17, 2026

Which status code is used by a REST API to indicate that the submitted payload is incorrect?

A 400 ✓ CORRECT

B 403

C 405

D 429

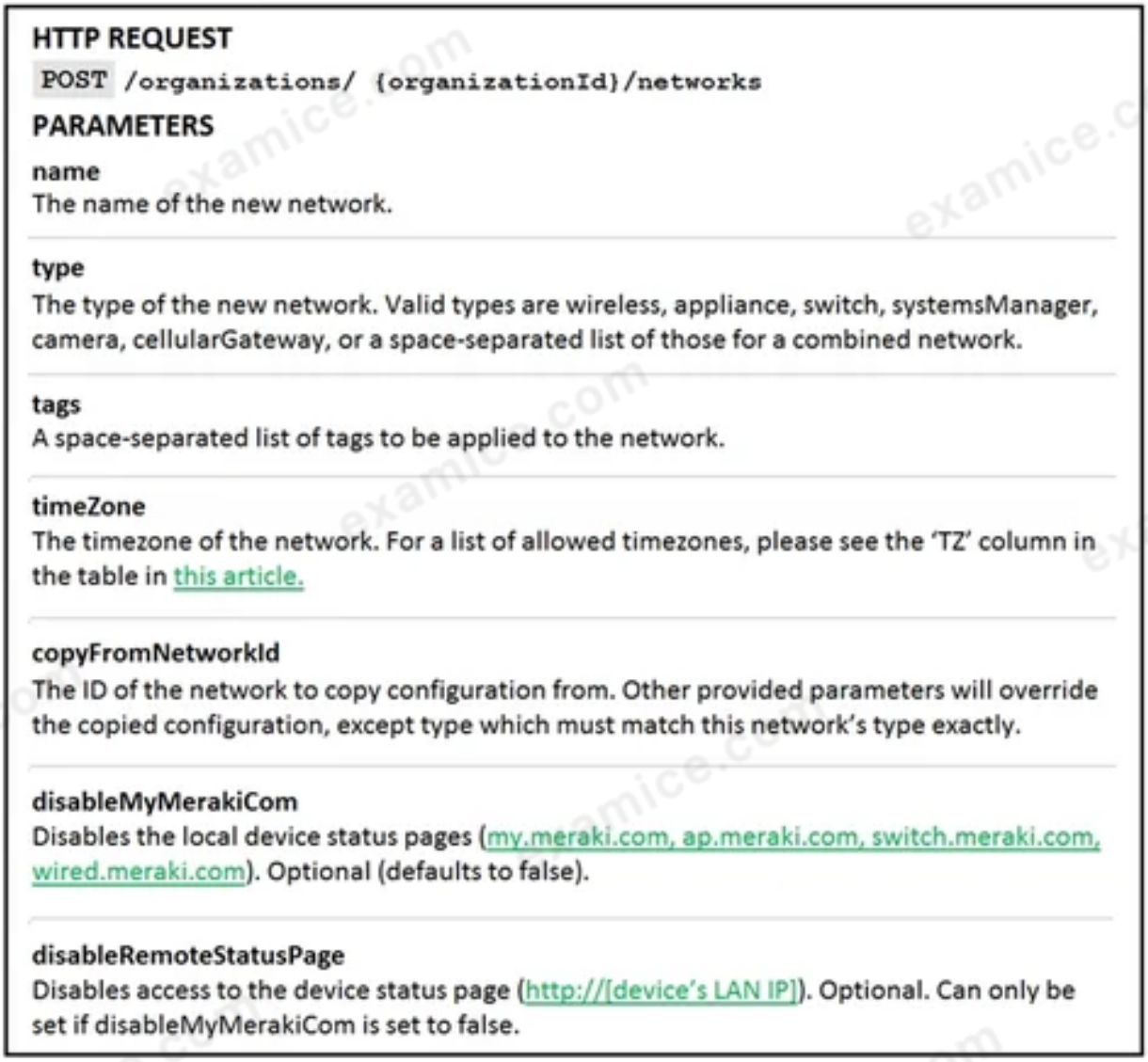
SUGGESTED ANSWER

The suggested answer is A.

The status code 400 is used by a REST API to indicate that the submitted payload is incorrect. This status code means 'Bad Request' and it generally implies that the server cannot or will not process the request due to a client error (e.g., malformed request syntax, invalid request message framing, or deceptive request routing).

DRAG DROP -

Refer to the exhibits.



HTTP REQUEST

`POST /organizations/ {organizationId}/networks`

PARAMETERS

name
The name of the new network.

type
The type of the new network. Valid types are wireless, appliance, switch, systemsManager, camera, cellularGateway, or a space-separated list of those for a combined network.

tags
A space-separated list of tags to be applied to the network.

timeZone
The timezone of the network. For a list of allowed timezones, please see the 'TZ' column in the table in this article.

copyFromNetworkId
The ID of the network to copy configuration from. Other provided parameters will override the copied configuration, except type which must match this network's type exactly.

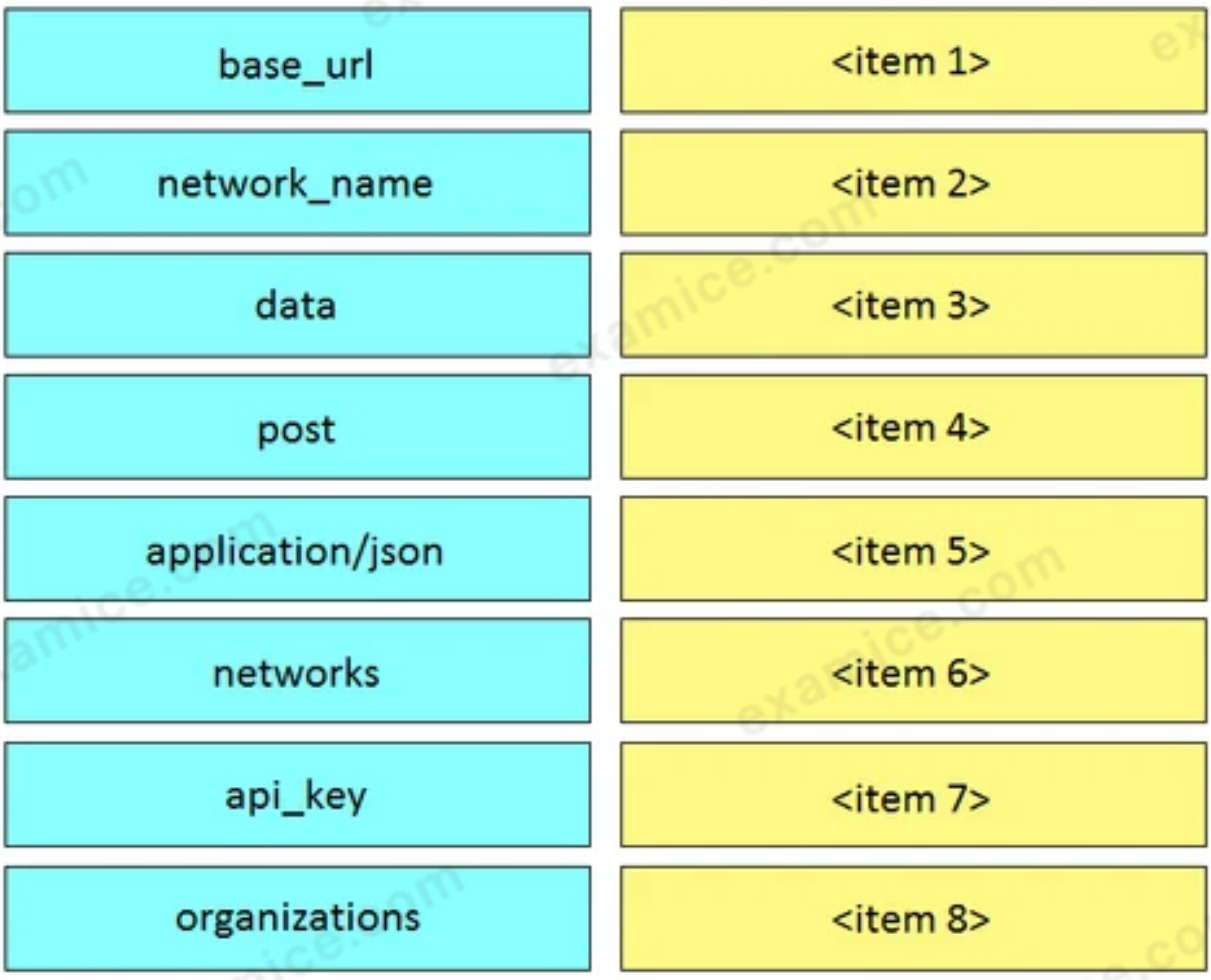
disableMyMerakiCom
Disables the local device status pages (my.meraki.com, ap.meraki.com, switch.meraki.com, wired.meraki.com). Optional (defaults to false).

disableRemoteStatusPage
Disables access to the device status page (http://[device's LAN IP]). Optional. Can only be set if disableMyMerakiCom is set to false.

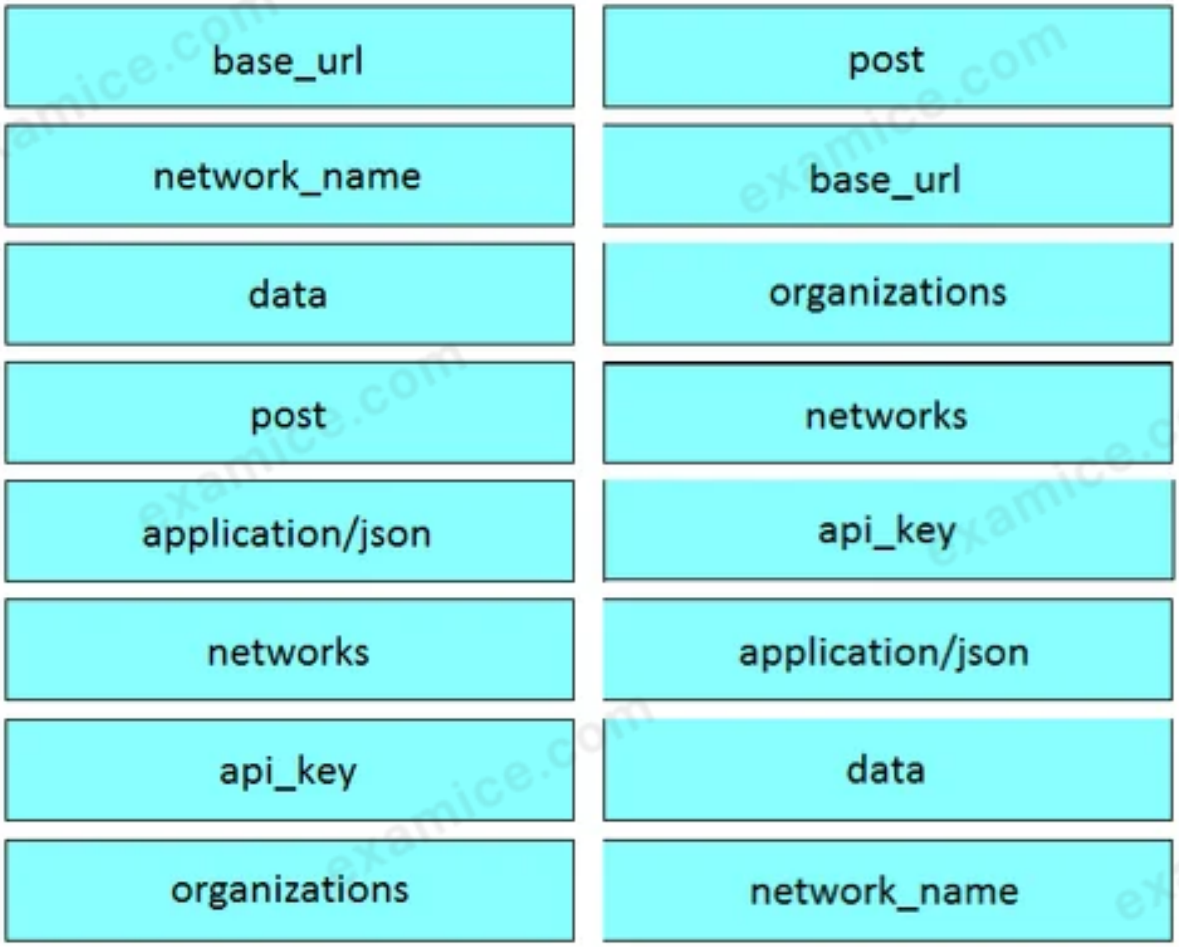
```
base_url = https://api.meraki.com/api/v0
org_id = 123456789
api_key = 1098hadpfsiapsf8ahf8ohp
network_name = "New Network"
requests.<item 1> (
    <item 2> + "/<item 3>/" + org_id + "/<item 4>",
    headers = {
        "X-Cisco-Meraki-API-Key": "<item 5>",
        "Content-Type": "<item 6>"
    },
    <item 7>=json.dumps ({ "name": <item 8>,
                           "type": "wireless switch"
                         }))
```

Drag and drop the code from the left onto the item numbers on the right to complete the Meraki Python script shown in the exhibit.

Select and Place:



SUGGESTED ANSWER



Personal copy · kamjunhao@gmail.com

Issued to kamjunhao@gmail.com · October 17, 2026

An application calls a REST API and expects a result set of more than 550 records, but each time the call is made, only 25 are returned. Which feature limits the amount of data that is returned by the API?

A pagination ✓ CORRECT

B payload limit

C service timeouts

D rate limiting

SUGGESTED ANSWER

The suggested answer is A.

When an application calls a REST API and receives a subset of the expected results, it is often due to pagination. Pagination is a feature used to divide a large dataset into smaller, manageable chunks, returning a fixed number of records per call. In this case, the REST API is returning 25 records per call, indicating that the data is being paginated.

Issued to kamjunhao@gmail.com · October 17, 2026

A developer is writing an application that uses a REST API and the application requires a valid response from the API. Which element of the response is used in the conditional check?

A body

B headers

C link

D URL

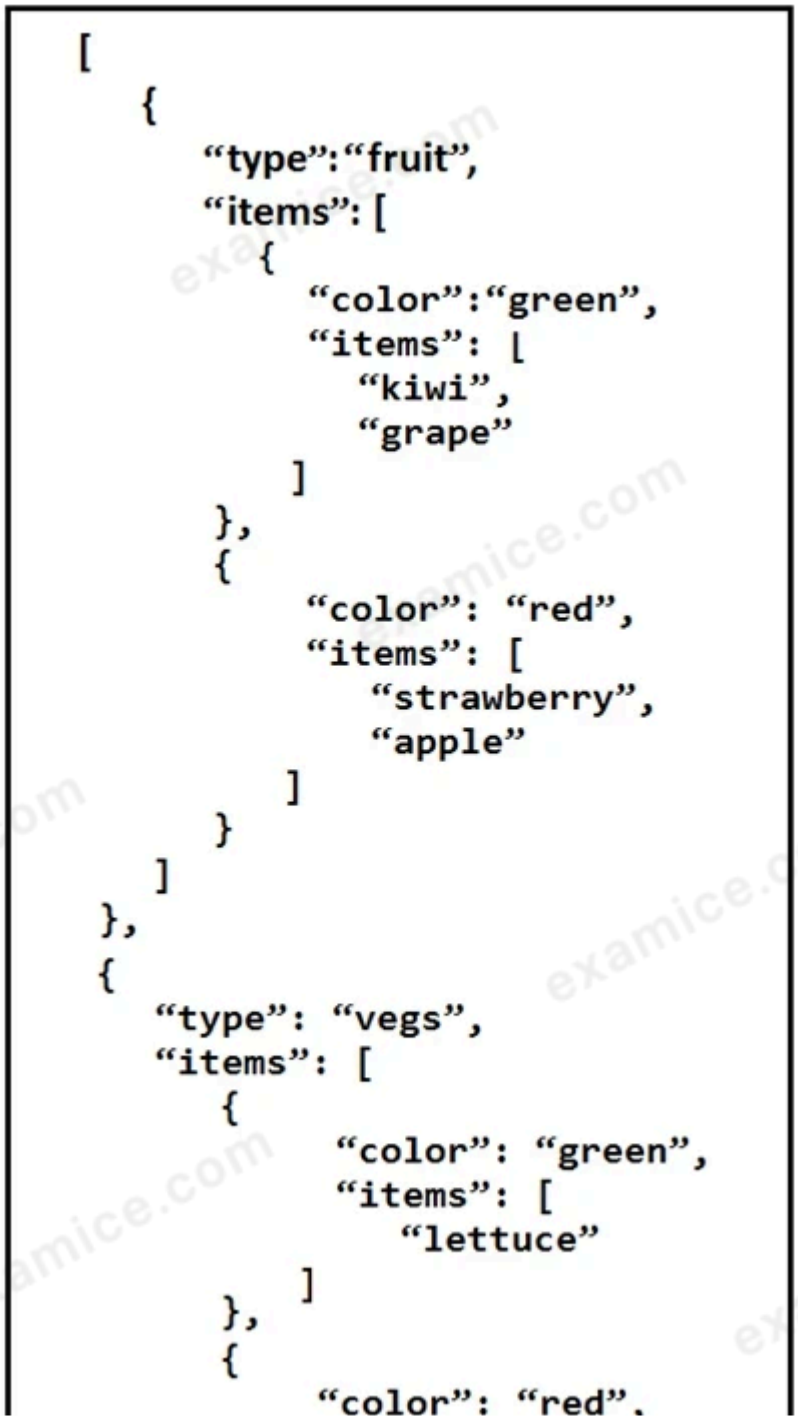
E status code ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is E.

When a developer wants to determine if a REST API response is valid, the status code is typically used in the conditional check. The status code, which is part of the response, indicates whether the request was successful (e.g., 200 OK for a successful GET request) or if there was an error (e.g., 404 Not Found, 500 Internal Server Error). This allows the application to intelligently handle different outcomes from the API call and take appropriate action based on the specific status code received.

Refer to the exhibit.



```
[
   {
      "type":"fruit",
      "items": [
         {
            "color":"green",
            "items": [
              "kiwi",
              "grape"
            ]
         },
         {
            "color": "red",
            "items": [
               "strawberry",
               "apple"
            ]
         }
   ]
},
{
   "type": "vegs",
   "items": [
      {
         "color": "green",
         "items": [
            "lettuce"
         ]
      },
      {
         "color": "red",
```

A REST API returns this JSON output for a GET HTTP request, which has been assigned to a variable called `vegetables`. Using Python, which output is the result of this command?
`print(filter(lambda 1: 1['type'] == 'fruit', vegetables) [0]['items'][0]['items'][0])`

A {'color': 'green', 'items': ['kiwi', 'grape']}

B ['kiwi', 'grape']

C lettuce

D kiwi ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

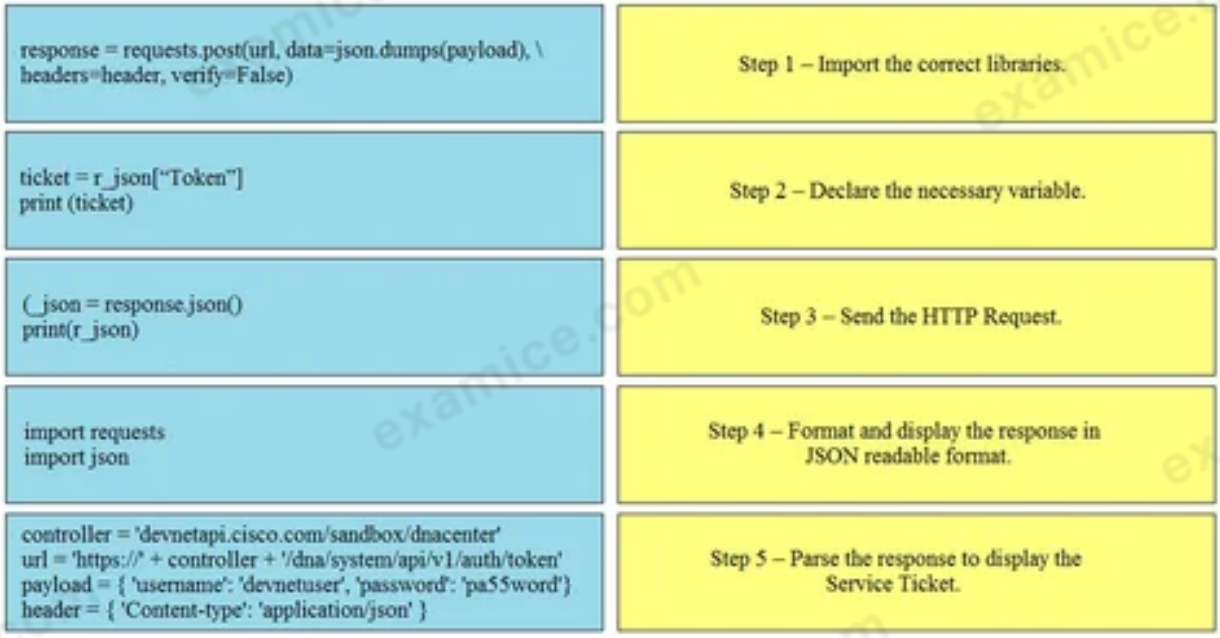
The Python command given first filters the list of dictionaries to find the dictionary where the 'type' is 'fruit'. This results in a list of dictionaries where 'type' is 'fruit'. The '[0]' index then selects the first dictionary in this filtered list. Inside this dictionary, 'items' is a key that maps to a list of dictionaries, so the '[0]' index selects the first dictionary in this list. This dictionary has another 'items' key which maps to a list of strings, and '[0]' selects the first string in this list, which is 'kiwi'. Hence, the result of the command is 'kiwi'.

Personal copy · kamjunhao@gmail.com

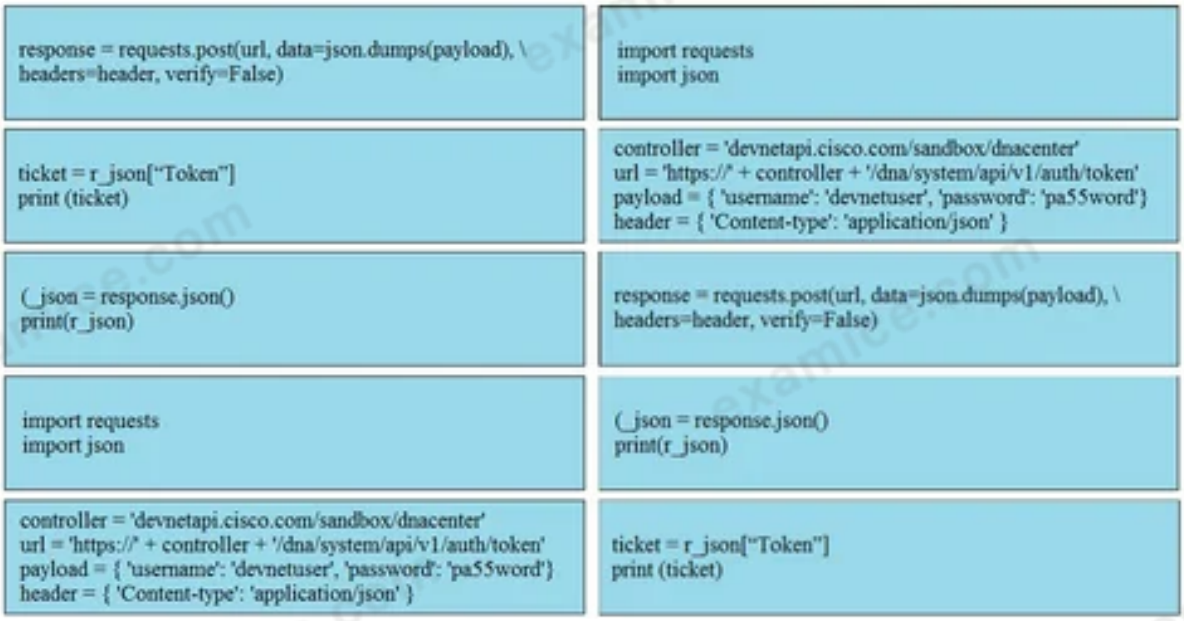
DRAG DROP -

Drag and drop the Python code from the left onto the correct step on the right to call a REST API.

Select and Place:



SUGGESTED ANSWER



A 401 HTTP response code is returned when calling a REST API. What is the error state identified by this response code?

A The server cannot process the request as it has detected an issue in the request syntax or body.

B The request has not been accepted because it requires authentication. ✓ CORRECT

C The sever accepted the request but the client is not authorized for this content.

D The server cannot find the requested resource because the path specified is incorrect.

SUGGESTED ANSWER

The suggested answer is B.

A 401 HTTP response code indicates that the request has not been applied because it lacks valid authentication credentials for the target resource. This means that the client must authenticate itself to get the requested response. This status code is typically returned when the credentials are either missing or incorrect, which results in the server rejecting the request. Hence, the correct error state identified by this response code is that the request requires authentication.

A developer is creating a script to interact with a REST API service which requires basic authentication. The credentials are "devnet:391665405" and the Base64 encoding of the credentials is "GV2bmV0dXNlcjpDaXNj=". Which payload and header combination must be used for authentication?

A.

```
payload = {
    'Authorization' : 'Basic GV2bmV0dXNlcjpDaXNj=',
    'Content-Type' : 'application/json'
}
headers= {}
```

B.

```
payload = {}
headers= {
    'Authorization' : 'Basic GV2bmV0dXNlcjpDaXNj=',
    'Content-Type' : 'application/json'
}
```

C.

```
payload = {
    'Authorization' : 'Bearer GV2bmV0dXNlcjpDaXNj=',
    'Content-Type' : 'application/json'
}
headers= {}
```

D.

```
payload = {}
headers= {
    'Authorization' : 'Bearer GV2bmV0dXNlcjpDaXNj=',
    'Content-Type' : 'application/json'
}
```

SUGGESTED ANSWER

To interact with a REST API service using basic authentication, the standard practice is to include the credentials in the request headers rather than the payload. Basic authentication requires an 'Authorization' header field with the value 'Basic ' followed by the base64-encoded string of 'username:password'. The provided base64 encoded credentials should be included in this header. 'Content-Type' is typically also specified in the headers to indicate the format of the payload data. Given these points, the correct payload and header combination is as follows: Payload should be empty ({}), and headers should include the Authorization and Content-Type fields as shown in Option B.

QUESTION 54

A developer needs a list of clients connected to a specific device in a Meraki network. After making a REST API call, the developer receives an unfamiliar response code. Which Cisco DevNet resource should be used to identify the meaning of the response code?

A API documentation ✓ CORRECT

B Code Exchange

C Learning Labs

D Sandbox

SUGGESTED ANSWER

The suggested answer is A.

The API documentation provides detailed information about the REST API, including the meaning of response codes. It is the appropriate resource to identify and understand unfamiliar response codes encountered while making API calls.

Issued to kamjunhao@gmail.com · October 17, 2026

A developer is trying to retrieve data over a REST API. The API server responds with an HTTP client error response code. After investigating the response, the developer realizes the response has a **Retry-After** header. What is the root cause of this error?

A An appliance limited the rate of requests to the transport layer.

B The REST service is unreachable at the time of the REST request.

C Too many requests were sent to the REST service in a given amount of time. ✓ CORRECT

D An appliance limited the rate of requests to the application layer.

SUGGESTED ANSWER

The suggested answer is C.

The root cause of this error is that too many requests were sent to the REST service in a given amount of time. The Retry-After header is typically used in conjunction with an HTTP 429 status, which indicates that the user has sent too many requests in a given amount of time and should wait before making more requests.

Refer to the exhibit.

Local File Attachments

To send local file attachments, simply post a message by including your access token in the Authorization header and the path to your local file with the files parameter. Optionally, you can also include a plain-text message with the attachment by using the text parameter. When uploading files directly from your local filesystem, your request will need to be a multipart/form-data request rather than JSON

```
curl --request POST \
  --form "files=@/home/desktop/example.png;type=image/png" \
  --form "roomId=Y21zY2....." \
  --form "text=example attached" \
https://webexapis.com/v1/messages
```

A developer needs to upload a local file by using the REST API. The developer gathers information according to the documentation and sends the request by using the cURL command in the exhibit but gets an error code. Which action should be followed to get a valid response?

A change content-type as JSON

B add the authorization header with the access token ✓ CORRECT

C and a username-password combination to request command

D change request method as GET

SUGGESTED ANSWER

The suggested answer is B.

To upload a local file using the REST API, you need to include your access token in the Authorization header. This step is necessary to authenticate the request and ensure it is authorized to make changes. Without the access token, the server will not recognize the request as valid, resulting in an error. Therefore, adding the authorization header with the access token is essential to resolving the error and obtaining a valid response.

DRAG DROP -

Refer to the exhibit.

Drag and drop the descriptors from the left onto the corresponding parts of the API request and response on the right.

Select and Place:

SUGGESTED ANSWER

Personal copy · kamjunhao@gmail.com

Issued to kamjunhao@gmail.com · October 17, 2026

Which action resolves a 401 error in response to an HTTP GET that is issued to retrieve a configuration statement using RESTCONF on a CSR 1000V?

A Change the HTTP method to PUT.

B Check the MIME types in the HTTP headers.

C Change the transport protocol to HTTPS.

D Check the authentication credentials. ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

A 401 Unauthorized error in response to an HTTP GET request indicates that the client must authenticate itself to get the requested response. To resolve this, the appropriate action is to check the authentication credentials. If the provided credentials are incorrect or missing, updating them should resolve the 401 error.

Issued to kamjunhao@gmail.com · October 17, 2026

Refer to the exhibit.

An API call is constructed to retrieve the inventory in XML format by using the API. The response to the call is 401 Unauthorized. Which two headers must be added to the API call? (Choose two.)

A Bearer-Token: dXNlcm5hbWU6cGFzc3dvcmQ=

B Content-Type: application/xml

C Authentication: Bearer dXNlcm5hbWU6cGFzc3dvcmQ=

D Accept: application/xml ✓ CORRECT

E Authorization: Bearer dXNlcm5hbWU6cGFzc3dvcmQ= ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D, E.

To address the 401 Unauthorized response from the API call, two headers need to be added. First, the 'Accept' header with 'application/xml' specifies the expected response format, which is necessary to indicate that the response should be in XML format. Second, the 'Authorization' header with 'Bearer dXNlcm5hbWU6cGFzc3dvcmQ=' is required to provide the correct authentication token, as the API documentation specifies the use of a Bearer token for authentication and authorization.

QUESTION 60

Issued to kamjunhao@gmail.com · October 17, 2026

Which HTTP response status code means `Forbidden`?

A 500

B 401

C 403 ✓ CORRECT

D 502

SUGGESTED ANSWER

The suggested answer is C.

The HTTP response status code 403 means 'Forbidden.' This status code indicates that the server understands the request but refuses to authorize it. Common reasons for this status code include insufficient permissions or access control settings preventing access to the requested resource.

Issued to kamjunhao@gmail.com · October 17, 2026

Which HTTP response code should be returned when the request is completed, but there is no content to send for the request?

A 100

B 204 ✓ CORRECT

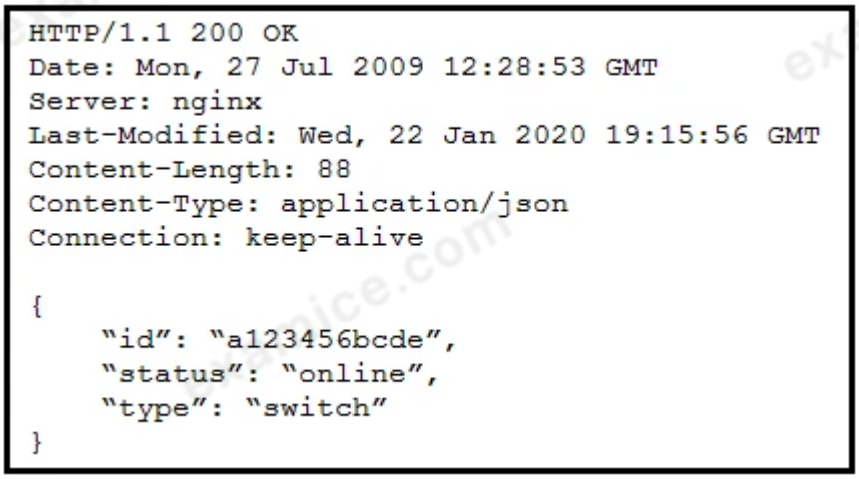
C 302

D 402

SUGGESTED ANSWER

The suggested answer is B.

The appropriate HTTP response code when the request is completed but there is no content to send for the request is 204. This status code indicates that the request was successful, but the server does not need to return a response body.



```
HTTP/1.1 200 OK
Date: Mon, 27 Jul 2009 12:28:53 GMT
Server: nginx
Last-Modified: Wed, 22 Jan 2020 19:15:56 GMT
Content-Length: 88
Content-Type: application/json
Connection: keep-alive

{
    "id": "a123456bcde",
    "status": "online",
    "type": "switch"
}
```

Refer to the exhibit. Which data is specified in the response header?

A Content-Type ✓ CORRECT

B HTTP/1.1 200 OK

C ג€typeג :€ג€switchג€

D { ג€idג :€ג€a123456bcdeג€, HTTP/1.1 200 OK ג€statusג :€ג€onlineג ,€ג€type:€ג ג€switchג€ }

SUGGESTED ANSWER

The suggested answer is A.

The data specified in the response header is the Content-Type, which is 'application/json'. This specifies the format of the data in the body of the response.

What are two use cases where webhooks are effective? (Choose two.)

A Filter out information from a response to an API call.

B Close a session with a web server after a specific amount of time.

C Inform a previously defined chat channel after a deployment fails. ✓ CORRECT

D Send an email to a customer of an online store after payment is complete. ✓ CORRECT

E Change the response format or content type of an API call.

SUGGESTED ANSWER

The suggested answer is C, D.

Webhooks are particularly effective for scenarios where immediate notifications are needed in response to specific events. Sending an email to a customer of an online store after payment is complete and informing a previously defined chat channel after a deployment fails are both events that happen asynchronously and benefit significantly from webhooks. These scenarios involve automatic triggers based on certain actions, making webhooks the ideal solution.

examice.com

Issued to kamjunhao@gmail.com · October 17, 2026

In which two ways is an application characterized when interacting with a webhook? (Choose two.)

A receiver ✓ CORRECT

B transaction monitor

C codec

D processor

E listener ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is A, E.

When an application interacts with a webhook, it can be characterized as a receiver because it receives data or payloads sent by the webhook. Additionally, it can be characterized as a listener because it listens for incoming HTTP requests from the webhook, indicating it must be ready to handle and process the information being sent. Therefore, the correct options are receiver and listener.

Issued to kamjunhao@gmail.com · October 17, 2026

How does requesting a synchronous API operation differ from requesting an asynchronous API operation?

A clients receive responses with a task ID for further processing

B clients subscribe to a webhook for operation results

C clients poll for the status of the execution of operations

D clients can access the results immediately ✓ CORRECT

SUGGESTED ANSWER

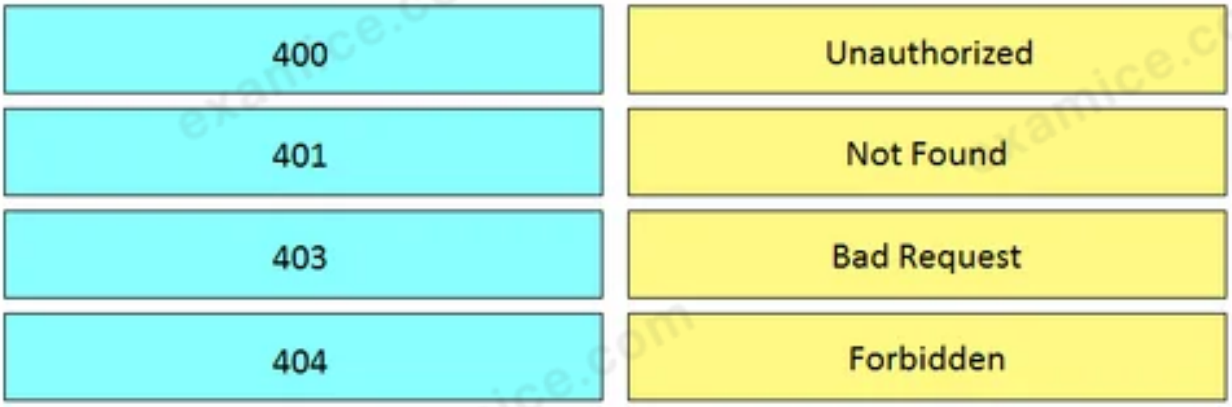
The suggested answer is D.

Requesting a synchronous API operation means that the client will receive the results immediately after making the request. This is because synchronous operations wait for the operation to complete and return the result before proceeding. In contrast, asynchronous API operations allow the client to continue other tasks while waiting for the operation to complete and typically notify the client when the result is ready.

DRAG DROP -

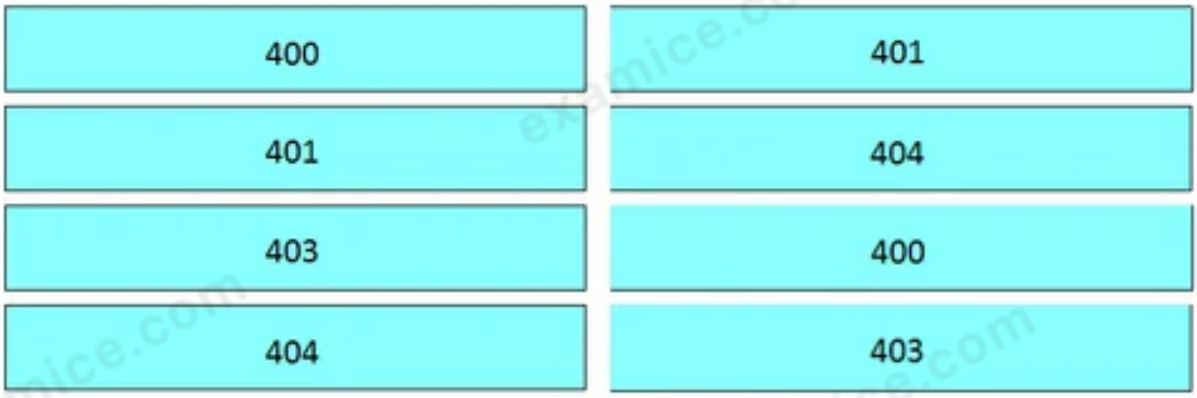
Drag and drop the HTTP status codes from the left onto the correct descriptions on the right.

Select and Place:



SUGGESTED ANSWER

Reference:
https://www.cisco.com/c/en/us/support/docs/security/web-security-appliance/118217-troubleshoot-wsa-00.html



Issued to kamjunhao@gmail.com · October 17, 2026

Which method should be used to safely store the API keys?

A in an environment variable unique to the system database

B encrypted in a configuration file that is separate from the code ✓ CORRECT

C plain text in the code as a constant

D encrypted in the code in a function

SUGGESTED ANSWER

The suggested answer is B.

To safely store API keys, the best practice is to encrypt the keys in a configuration file that is separate from the code. This approach ensures that the keys are not exposed within the codebase and are protected through encryption. Storing keys in plain text within the code (option C) or even encrypted in the code (option D) increases the risk of exposure if the code is compromised. While using environmental variables (option A) is a common practice, it does not inherently provide encryption and can be less secure in certain environments compared to an encrypted, separate configuration file.

FILL BLANK -

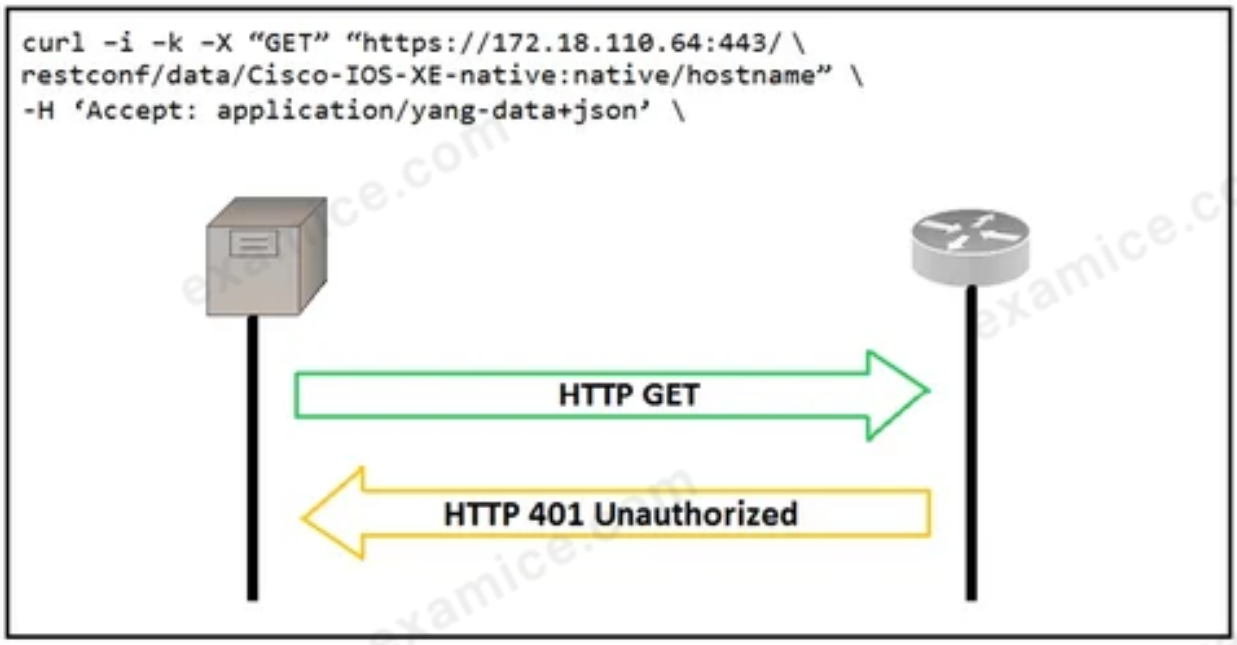
Fill in the blanks to complete the statement.

Given a username of `devnet` and a password of `cisco123`, applications must create a base64 encoding of the string `_____` when sending HTTP requests to an API that uses `_____` authentication.

SUGGESTED ANSWER

Given a username of devnet and a password of cisco123, applications must create a base64 encoding of the string devnet:cisco123 when sending HTTP requests to an API that uses Basic authentication. The string 'devnet:cisco123' must be base64 encoded as it is the standard format for Basic authentication. With HTTP Basic authentication, the username and password are concatenated with a colon (:) and then encoded using base64. This encoded string is then sent in the Authorization header of the HTTP request.

Refer to the exhibit.



An administrator attempts to perform a GET using the Cisco IOS XE RESTCONF API to return the hostname of a device. The sequence diagram illustrates theHTTP messages observed. Which change to the API request resolves the issue?

A Remove the ג€"H 'Accept: application/yang-data+json' HTTP header because it is not required.

B Add ג€"'u cisco:cisco' in the end of the cURL command ✓ CORRECT

C Change the request method from ג€"X ג€GETג€ to ג€"X ג€POSTג€

D Add Content-Type HTTP header with 'application/yang-data+json' using ג€"H 'Content-Type: application/yang-data+json'

SUGGESTED ANSWER

The suggested answer is B.

The HTTP 401 Unauthorized status code indicates that authentication is required and has failed or not been provided properly. The cURL command in the exhibit is missing the authentication credentials. Adding the '-u cisco:cisco' to the cURL command will provide the

necessary authentication credentials in the format of username:password, which is required for accessing RESTCONF APIs on Cisco IOS XE devices.

QUESTION 70

What are two benefits of managing network configuration via APIs? (Choose two.)

A more security due to locking out manual device configuration

B configuration on devices becomes less complex

C eliminates the need of legacy management protocols like SNMP

D reduction in network changes performed manually ✓ CORRECT

E increased scalability and consistency of network changes ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D, E.

Managing network configuration via APIs brings about two significant benefits. First, it reduces the number of network changes performed manually by automating the process, which minimizes human error and increases efficiency. Second, it increases scalability and consistency of network changes, as APIs can ensure uniform configuration across multiple devices and can handle a large number of devices simultaneously. These benefits help in maintaining a more robust and efficient network management system.

FILL BLANK -

Fill in the blanks to complete the cURL command that invokes a RESTful API to retrieve a resource in JSON format using OAuth. curl "X _____ -H ` _____: application/json` \

-H ` _____: Bearer AbCdEf123456` https://localhost/api/myresource

SUGGESTED ANSWER

To fill in the blanks for the cURL command that invokes a RESTful API to retrieve a resource in JSON format using OAuth, the following components are required: the HTTP method, the 'Accept' header, and the 'Authorization' header. The correct and complete cURL command would be 'curl -X GET -H 'Accept: application/json' -H 'Authorization: Bearer AbCdEf123456' https://localhost/api/myresource'. The HTTP method 'GET' is appropriate for retrieving resources, the 'Accept' header specifies the expected response format, which is 'application/json', and the 'Authorization' header includes the OAuth Bearer Token for authentication.

Issued to kamjunhao@gmail.com · October 17, 2026

Which Cisco DevNet resource allows access to products in a development lab to explore, learn, and build applications that use Cisco APIs?

A DevNet Code Exchange

B DevNet Sandbox ✓ CORRECT

C DevNet Communities

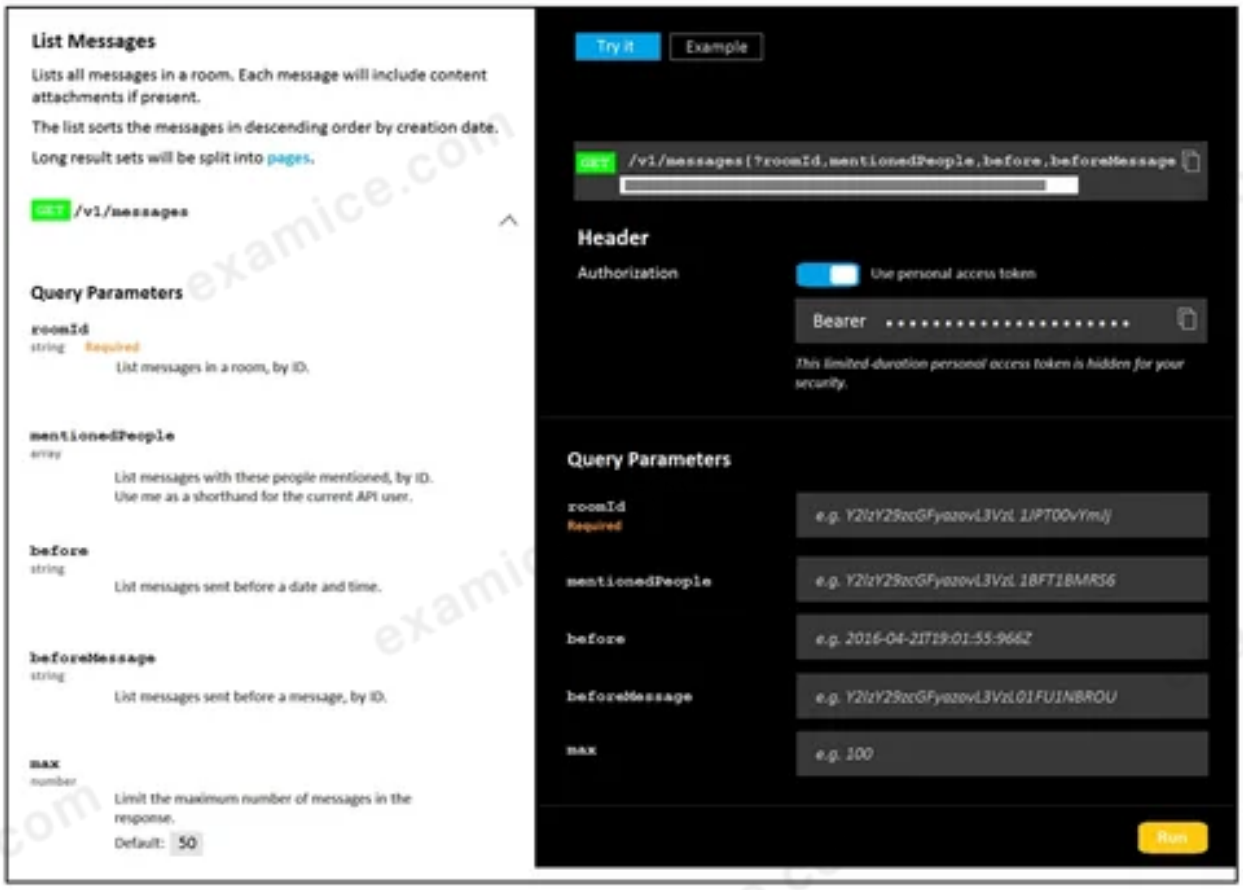
D DevNet Automation Exchange

SUGGESTED ANSWER

The suggested answer is B.

DevNet Sandbox is the Cisco DevNet resource that allows users to access products in a development lab environment to explore, learn, and build applications using Cisco APIs. The sandbox provides a safe and controlled space for developers and other users to experiment with Cisco technologies without the need to set up their own physical hardware or software environments.

Refer to the exhibits.




```
bash-3.2$ curl -H "Content-Type: application/json" -H "Authorization:
Fj2zzykEa091ic9GK2j8LtElHklH6oRHPQdwlpAT60I7NDThhNwZl2B5pqMg14kK_b9ei59ISAClY7-
NarA-2n9H-tGgt-SxQ39iDejgcs" -i -X POST -d "{ \"roomId\":
\"Y2geK53sjEknosrC7SwQ5ZGL99pHgiuScB7DfNvUsr8Xx4wDKLiPORcEkryAnM3QmK9LQZsP0G4\"
, \"text\": \"test2\" }" https://api.ciscospark.com/v1/messages

HTTP/1.1 401 Unauthorized
Via: 1.1 linkerd
TrackingID: ROUTER_5E0FDC8C-EB6F-01BB-00EF-806BF1BD00EF
Date: Sat, 04 Jan 2020 00:30:04 GMT
Server: Redacted
Content-Length: 267
Content-Type: application/json
Strict-Transport-Security: max-age=63072000; includeSubDomains; preload

{"message":"The request requires a valid access token set in the Authorization request
header.","errors":[{"description":"The request requires a valid access token set in the
Authorization request
header."}],"trackingId":"ROUTER_5E0FDC8C-EB6F-01BB-00EF-806BF1BD00EF"}

bash-3.2$
```

A developer is troubleshooting an API with the given API documentation and cURL command. What is the cause of this problem?

A The authorization header is missing or incomplete. ✓ CORRECT

B The request body is missing or incomplete.

C The API token specified is expired.

D The user is not allowed to post messages from their account.

SUGGESTED ANSWER

The suggested answer is A.

The error message explicitly states 'The request requires a valid access token set in the Authorization request header.' The most likely cause of the 401 Unauthorized error is that the 'Bearer' keyword is missing from the authorization header. Including 'Bearer' before the token is a common requirement in API authorization processes and is likely the missing element causing the request to fail.

Personal copy · kamjunhao@gmail.com

Which two statements describe the traits of an asynchronous API call? (Choose two.)

A Code execution blocks or waits for the call to an API to return.

B A callback function typically is used to process the response from an API call. ✓ CORRECT

C A call to an API does not block the code, but rather it allows application processing to continue. ✓ CORRECT

D The end user can experience latency or performance lag while waiting for the API call to return.

E The order in which API calls return can be guaranteed.

SUGGESTED ANSWER

The suggested answer is B, C.

An asynchronous API call does not block the code execution but allows the application processing to continue, making it non-blocking. This is a defining characteristic of asynchronous calls. Additionally, a callback function is typically used to process the response from an asynchronous API call, allowing the application to handle the response once it becomes available without having to wait for it. These traits highlight the key differences between synchronous and asynchronous processing.

Which mechanism is used to consume a RESTful API design when large amounts of data are returned?

A data sets

B scrolling

C pagination ✓ CORRECT

D blobs

SUGGESTED ANSWER

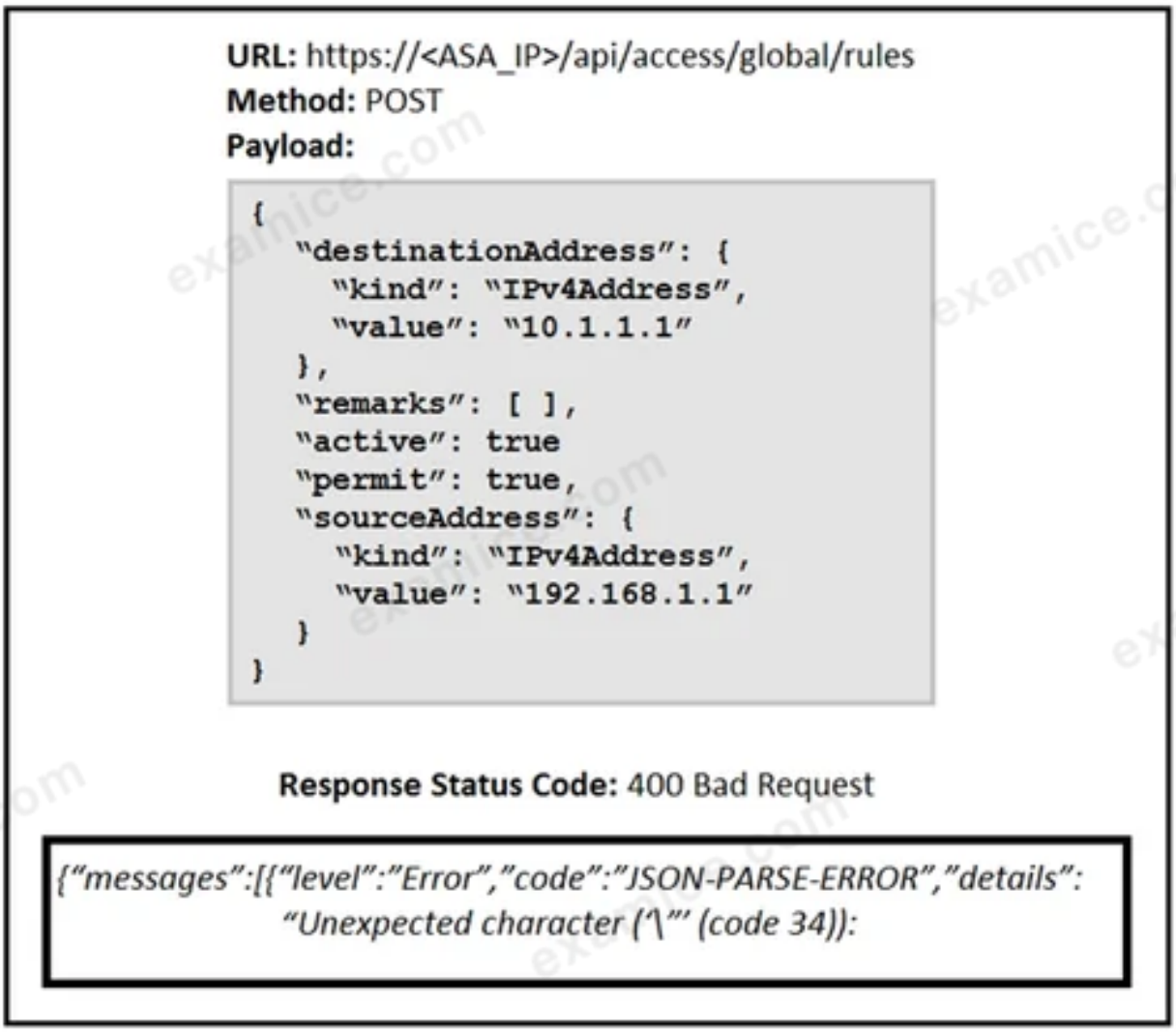
The suggested answer is C.

The mechanism used to consume a RESTful API design when large amounts of data are returned is pagination. Pagination allows the data to be returned in smaller, manageable chunks or 'pages', thus making it more efficient and easier to handle large datasets without overwhelming the client or server.

examice.com

Issued to kamjunhao@gmail.com · October 17, 2026

Refer to the exhibit.



What caused the error in this API request?

A The API resource does not support the POST operation.

B The submitted JSON payload has a formatting issue. ✓ CORRECT

C The API resource does not support JSON format payloads.

D The submitted JSON payload includes a field that is not supported by the API resource.

SUGGESTED ANSWER

The suggested answer is B.

The error in the API request was caused by a formatting issue in the JSON payload. Specifically, there is a missing comma after the "active": true field, which disrupts the proper structure of the JSON format. JSON data structures require commas to separate key-value pairs, and the absence of a comma creates a syntax error, triggering the JSON parse error indicated in the response.

QUESTION 77

Which two use cases are supported by Meraki APIs? (Choose two.)

A Build location-aware apps from Wi-Fi and LoRaWAN devices.

B Build a custom Captive Portal for Mobile Apps. ✓ CORRECT

C Configure network devices via the Dashboard API. ✓ CORRECT

D Deploy applications onto the devices.

E Retrieve live streams from a Meraki Camera.

SUGGESTED ANSWER

The suggested answer is B, C.

Meraki APIs support the following use cases: building a custom Captive Portal for Mobile Apps and configuring network devices via the Dashboard API. Through the Captive Portal API, organizations can create customized splash pages for device users, enhancing user engagement and experience. The Dashboard API allows programmatic configuration and management of Meraki network devices, enabling streamlined network administration and automated deployment of configuration settings.

Which API is used to obtain data about voicemail ports?

A Webex Teams

B Cisco Unified Communications Manager ✓ CORRECT

C Finesse Gadgets

D Webex Devices

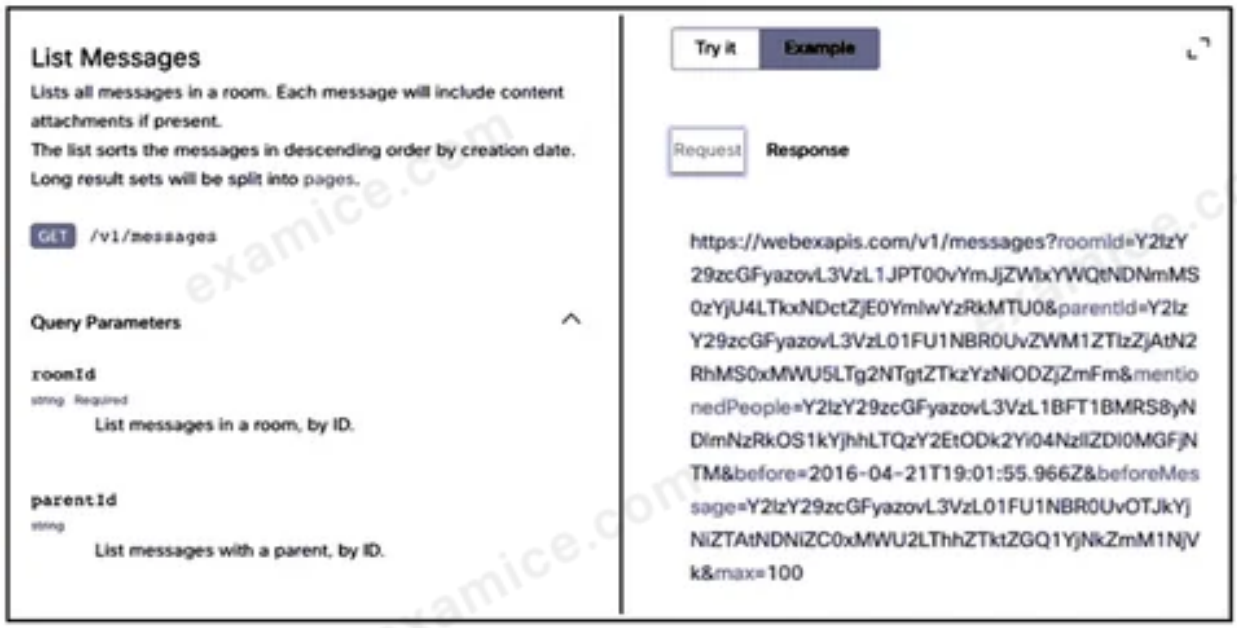
SUGGESTED ANSWER

The suggested answer is B.

Cisco Unified Communications Manager (CUCM) provides the necessary APIs to access and manage various components of a Cisco collaboration environment, including voicemail ports. Voicemail ports refer to the interfaces that allow voicemail components to connect and interact within the telephony system, and are typically managed within the CUCM framework.

examice.com

DRAG DROP -



Refer to the exhibit. A developer needs to automatically retrieve all of the messages of a Webex room with the roomId of`Y2lzY29zcGFyazovL3Vz397468502YjU5NjAtNTk0Zi0xMWVhLTk0Mj`. Using the Webex API documentation shown, drag and drop the code snippets from below onto the code to complete the Python script to list all of the messages in the room. Not all options are used.Select and Place:

SUGGESTED ANSWER

Answer Area

```
import requests

webex_token = "NDA2OGV...f0-4434-a696-84fee4047e0a"

room_id = "Y2lzY29zcGF...jU5NjAtNTk0Zi0xMWVhLTk0Mj"

url = "https://webexapis.com/v1/ [messages?] [roomId] =" + [room_id]

payload = {}
headers = {
  'Authorization': 'Bearer ' + [webex_token]
}

response = requests.request("GET", url, headers=headers,data = payload)

print(response.text.encode('utf8'))
```

[room]

[messages]

What is a difference between a synchronous API and an asynchronous API?

A Synchronous API calls require an authentication header to be sent while asynchronous calls do not require authentication.

B Synchronous API calls are returned immediately while asynchronous calls do not guarantee an immediate response. ✓ CORRECT

C An asynchronous API can make offline calls while synchronous APIs do not have this capability.

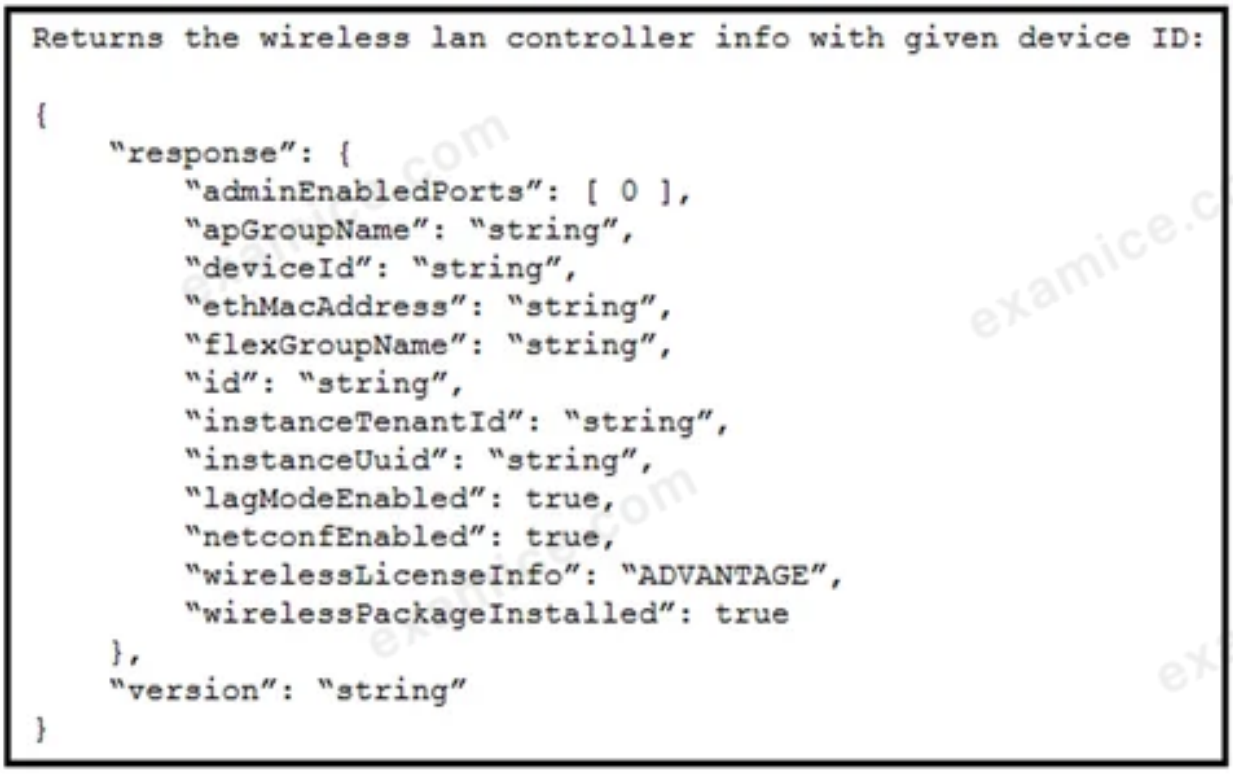
D An asynchronous API can make a larger number of calls in a specified time period than a synchronous API.

SUGGESTED ANSWER

The suggested answer is B.

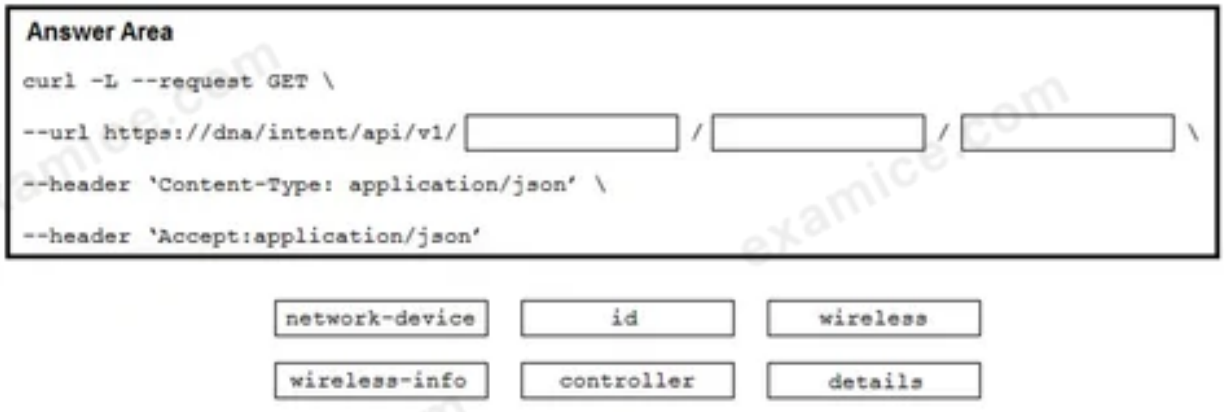
A synchronous API call is a blocking call that waits for the server to process the request and return a response. It must wait for the action to be completed before the response is delivered. In contrast, an asynchronous API call does not block the client. It returns immediately after the request is sent, allowing the client to continue processing other tasks while waiting for the actual response to be delivered later, potentially through a callback mechanism or by polling. This key difference lies in the blocking nature of synchronous calls and the non-blocking nature of asynchronous calls.

DRAG DROP -

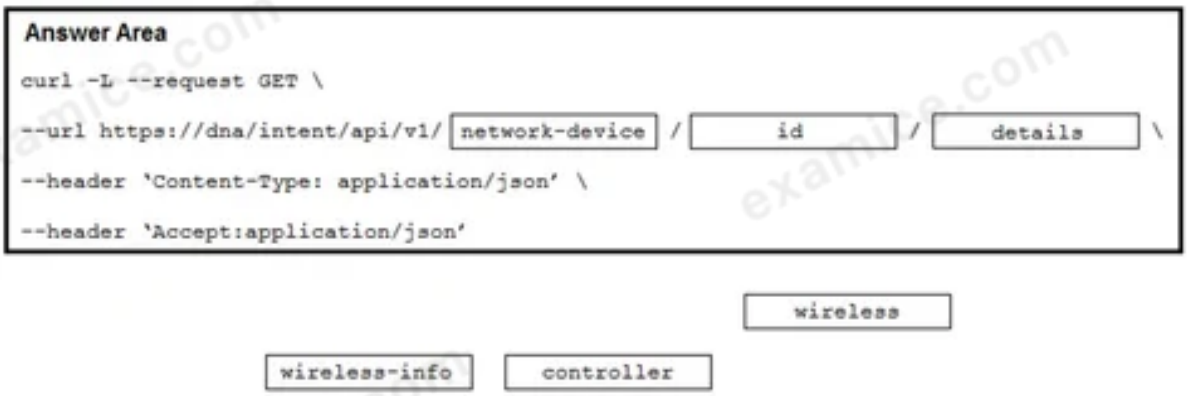


Refer to the exhibit. Drag and drop the code from the bottom onto the blanks in the code to construct a cURL command using the Cisco DNA Center API, which will provide the details of a WLAN controller with Id ab123456789. Not all options are used.

Select and Place:



SUGGESTED ANSWER



Personal copy · kamjunhao@gmail.com

```
import requests
from requests.auth import HTTPBasicAuth
headers = {'Content-Type': 'application/json'}
base_url = 'https://10.1.2.3'
auth_url = '/dna/system/api/v1/auth/token'
devices_url = '/dna/intent/api/v1/network-device'

token = requests.post(
   base_url + auth_url, headers=headers,
   auth=HTTPBasicAuth(username="admin", password="DevNet123"))
data = token.json()

query_string_params = {'platformId": "A1234-12B'}
response = requests.get(
    base_url + devices_url, headers=headers,
    params=query_string_params)
print(f'{response.code}')
```

Refer to the exhibit. A network engineer must manage the network devices. The engineer prepares a Python script to authenticate to the Cisco DNA Center API and request a device list. The device request fails and returns error code 401. Which action solves the problem?

A Update the credentials that are already supplied and retry the request.

B Send the request to another network in case there are reachability issues.

C Send the Authentication header in the request with a valid configuration. ✓ CORRECT

D Update the API URL, which matched the API endpoint for device list.

SUGGESTED ANSWER

The suggested answer is C.

The device request fails with the error code 401, which signifies 'Unauthorized' access. This indicates that the request to obtain the device list lacks proper authentication. The authentication token, which is acquired in the initial request (token request), should be included in the request header for subsequent API calls. Updating the headers to include the Authorization token in the request when obtaining the device list will authenticate the request and allow access to the device information.

Issued to kamjunhao@gmail.com · October 17, 2026

Code:

```
import requests

url = 'https://en.wikipedia.org/api/rest_v1/page/title/Cisco_Systems'
s = requests.Session()

def get_json(session=None, url="https://example.com"):
    try:
        if not session or type(session) is not requests.sessions.Session:
            raise Exception("Invalid session type")

        r = session.get(url)

        if r.status_code < 200 or r.status_code > 307:
            raise Exception("Status Code not within 2xx and 3xx")

        return r.json()

    except requests.exceptions.RequestException as e:
        raise e("Request failed")
    finally:
        session.close()

if __name__ == "__main__":
    items = get_json(session=s,url=url)
```

payload:

```
{'items': [{'title': 'Cisco_Systems', 'page_id': 51746, 'rev': 930604297,
'tid': '58b40b30-20e7-11ea-a7a9-f1850c7cfe85', 'namespace': 0, 'user_id':
19450869, 'user_text': 'Dhruvh
', 'timestamp':'2019-12-13T16:47:52Z', 'comment': '/* 1984-1995: Origins
and initial growth */', 'tags': [], 'restrictions': [], 'page_language':
'en', 'redirect': False}]}
```

Refer to the exhibit. Which command needs to be placed on the box where the code is missing to output the value of page_id in the Python 3.7 script?

A print(items.get('items')[0].get('page_id')) ✓ CORRECT

B print(items.get('items').get('page_id'))

C print(items['items']['page_id'].keys())

D print(items['items']['page_id'])

SUGGESTED ANSWER

The suggested answer is A.

> The payload shows that 'items' is a list containing dictionaries. To access the 'page_id' value, you first need to access the first element of the list using [0] and then get the 'page_id' from that dictionary. Therefore, the correct command to place in the box is print(items.get('items')[0].get('page_id')).

Personal copy · kamjunhao@gmail.com

Refer to the exhibit.

Generating Client ID and API Key:

- Log in to your AMP for Endpoints Console.
- Go to Accounts > Business Settings.
- Click Configure API Credentials under Features to generate the Client ID and secure API Key.

Once you have the API client ID and API key, you can make the API calls as follows:

```
https://<your_client_id>:<your_api_key>@<api_endpoint>
```

Alternatively you can use Basic HTTP Authentication. Base 64 encode the string ":", and send that prefixed with the string "Basic" as the authorization header. For instance, if your client_id was 1234, and your api_key was "atest", then it would be base64 encoded to "MTIzNDphdGVzdA==", and your header would be:

```
Authorization: Basic MTIzNDphdGVzdA==
```

The documentation outlines how to use credentials for the AMP API. Which script successfully performs an API call using basic authentication?

1. A.

```
import requests
amp_auth_token = 'a1b2c3d4e5f6g7h8i9j0 k1l2m3n4o5p6'
url = 'https://api.amp.cisco.com/v1/version'
request = requests.get(url,
   auth=(amp_auth_token))
print(request.json())
```

1. B.

```
import requests
amp_client_id = 'a1b2c3d4e5f6g7h8i9j0'
amp_api_key = 'a1b2c3d4-e5f6-g7h8-i9j0-k1l2m3n4o5p6'
url = 'https://api.amp.cisco.com/v1/version'
request = requests.get(url,
   auth=(amp_client_id, amp_api_key))
print(request.json())
```

1. C.

```
import requests
amp_client_id = 'a1b2c3d4e5f6g7h8i9j0'
amp_api_key = 'a1b2c3d4-e5f6-g7h8-i9j0-k1l2m3n4o5p6'
url = 'https://api.amp.cisco.com/v1/version'
request = requests.get(url))
print(request.json())
```

1. D.

```
import requests
amp_client_id = 'a1b2c3d4e5f6g7h8i9j0'
amp_login_password = '10g1n_p@ssw0rd'
url = 'https://api.amp.cisco.com/v1/version'
request = requests.get(url,
   auth=(amp_client_id, amp_login_password))
print(request.json())
```

SUGGESTED ANSWER

To successfully perform an API call using basic authentication for the AMP API, the correct script should include the use of both the client ID and API key. According to the exhibit provided, the authentication needs to be carried out using Basic HTTP Authentication by including a base64-encoded string in the header or by directly providing the credentials using the requests library's built-in authentication methods. The correct script in this context should specify the client ID and API key for basic authentication. The only script that aligns with this approach is Option B, which correctly provides the credentials `(amp_client_id, amp_api_key)` in the `auth` parameter of the `requests.get` method. Hence, Option B is the correct answer.

API Documentation:

Integrations are how you request permission to invoke the Webex REST API on behalf of another Webex Teams user. To do this in a secure way the API supports the OAuth 2 standard which allows third-party integrations to get a temporary access token for authenticating API calls instead of asking users for their password.

Script:

```
01   import requests
02
03   base_url = "https://webexapis.com/v1"
04   header = {"Key": "Value"}
05
06   url = "{}/{}".format(base_url, "/teams")
07   print(requests.get(url, headers=header).status_code)
```

Refer to the exhibit. A developer creates a Python script that queries Cisco Webex. When the script is executed, a 401 code is returned. After troubleshooting, the developer discovers that the service is missing privileges. Which change to the header in line 4 of the script results in the code 200?

A header = {ג€Authenticationג :€ג€Bearer YOUR_TOKENג€}

B header = {ג€Authentication Bearerג : €ג€YOUR_TOKENג€

C header = {ג€Authorization Bearerג : €ג€YOUR_TOKENג€}

D header = {ג€Authorizationג : €ג€Bearer YOUR_TOKENג€} ✓ CORRECT

SUGGESTED ANSWER

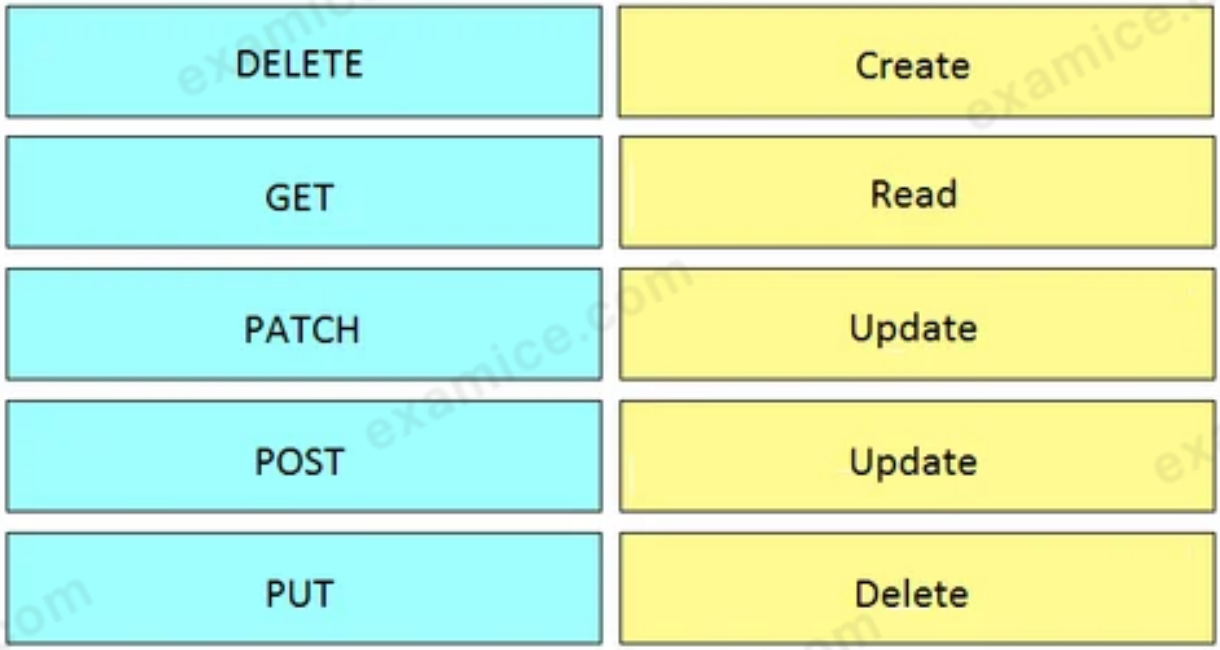
The suggested answer is D.

The correct format for the header in OAuth 2.0 authentication is 'Authorization: Bearer YOUR_TOKEN'. This header provides the necessary access token for authentication, which allows successful API calls to Webex. The script currently returns a 401 error code, indicating unauthorized access. By adjusting the header to use 'Authorization' instead of 'Authentication' and including the 'Bearer' prefix followed by the token, the script will be properly authenticated and should return a 200 status code.

DRAG DROP -

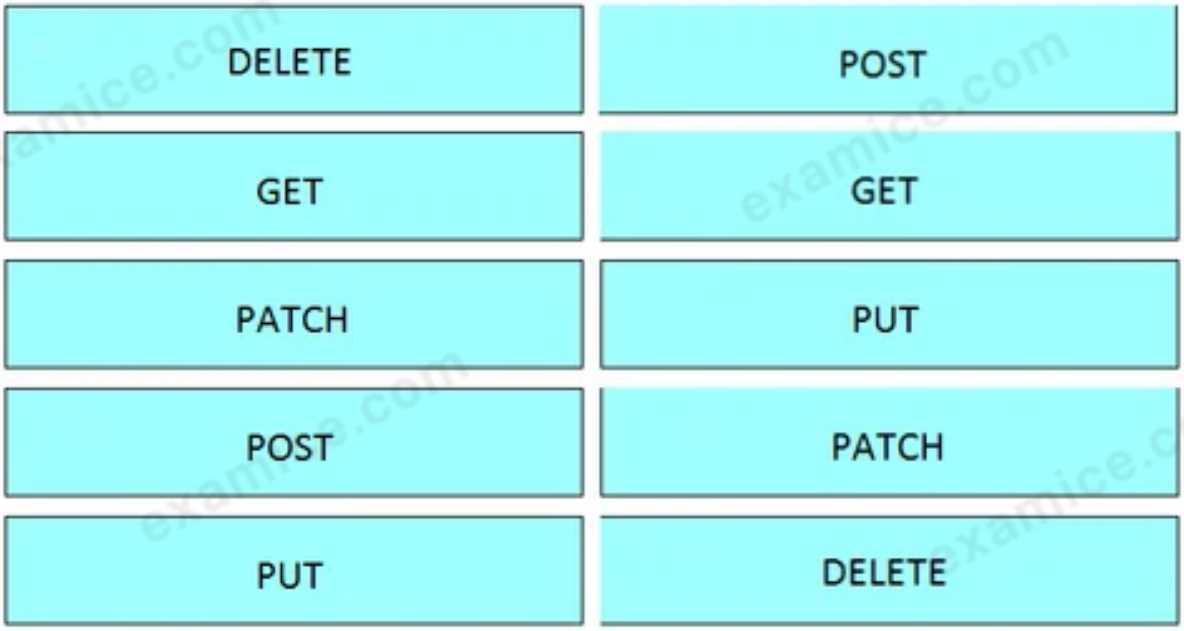
Drag and drop the HTTP methods from the left onto their generally accepted corresponding create, read, update, and delete operations on the right.

Select and Place:



SUGGESTED ANSWER

Reference:
https://www.cisco.com/c/en/us/td/docs/cloud-systems-management/application-policy-infrastructure-controller-enterprise-module/1-2-x/config-guide/ b_apic-em_config_guide_v_1-2-x/b_apic-em_config_guide_v_1-2-x_chapter_01001.pdf



Personal copy · kamjunhao@gmail.com

DRAG DROP -

Refer to the exhibit.

List the clients that have used this network in the timespan

HTTP REQUEST

`GET /networks/ {networkId}/clients`

PARAMETERS

t0

The beginning of the timespan for the data. The maximum lookback period is 31 days from today.

timespan

The timespan for which the information will be fetched. If specifying timespan, do not specify parameter t0. The value must be in seconds and be less than or equal to 31 days. The default is 1 day.

perPage

The number of entries per page returned. Acceptable range is 3 – 1000. Default is 10.

startingAfter

A token used by the server to indicate the start of the page. Often this is a timestamp or an ID but it is not limited to those. This parameter should not be defined by client applications. The link for the first, last, prev, or next page in the HTTP Link header should define it.

endingBefore

A token used by the server to indicate the end of the page. Often this is a timestamp or an ID but it is not limited to those. This parameter should not be defined by client applications. The link for the first, last, prev, or next page in the HTTP Link header should define it.

```
base_url = "https://api.meraki.com/api/v0"
network_id = "L_12345678910"
api_key = "1098hadpfsiapsf8ahf8ohp"
requests.<item 1> (
    <item 2> + "/<item 3>/" + <item 4>, "/<item 5>",
    headers = {
        "X-Cisco-Meraki-API-Key": <item 6> ,
        "Content-Type": "<item 7>"
    },
    <item 8>={"<item 9>":<item 10>})
```

Drag and drop the code from the left onto the item numbers on the right to complete the Meraki code to obtain a list of clients which have used this network.

Select and Place:

params	<item 1>
networks	<item 2>
timespan	<item 3>
get	<item 4>
432000	<item 5>
base_url	<item 6>
application/json	<item 7>
network_id	<item 8>
clients	<item 9>
api_key	<item 10>

SUGGESTED ANSWER

Personal copy · kamjunhao@gmail.com

In Python, which expression checks whether the script returns a success status code when the Requests library is used?

A response.status_code == requests.codes.ok ✓ CORRECT

B response.code == requests.codes.ok

C response.status_code == requests.ok

D response.status_code != requests.codes.ok

SUGGESTED ANSWER

The suggested answer is A.

In Python's Requests library, the expression to check if the response was successful (i.e., returned a status code of 200) is 'response.status_code == requests.codes.ok'. This checks if the status code of the response matches the code for a successful HTTP request, which is 200. The other options either use incorrect attributes or incorrect comparisons.

examice.com

```
import requests
from requests.auth import HTTPBasicAuth
from config import url, username, password
response = requests.login(url, auth=HTTPBasicAuth (username, password), verify=False)
print response.status_code
```

Refer to the exhibit. The script returns an output of 401. To use the supplied URL, an HTTP GET request must be sent with an Authorization header. The header value is a base64 encoded concatenation of the username and password. Which action must be taken to return an output of 200?

A Change the verify=False setting in the request to verify=True to enable HTTP Basic authentication.

B Verify that the username and password values imported from the configuration file are still valid.

C Insert an Authorization header with the values username:password from the supplied configuration file. ✓ CORRECT

D Modify the method from LOGIN to GET and supply the username:password value as JSON payload.

SUGGESTED ANSWER

The suggested answer is C.

To return an output of 200, the correct approach is to ensure the authorization header is correctly set. HTTP Basic Authentication requires sending the username and password encoded in Base64 format in the Authorization header. Using the 'requests' library, you can achieve this by including an authorization header with the username and password or employing HTTPBasicAuth. Thus, inserting an Authorization header with the values username:password from the supplied configuration file is the proper action.

```
>>> response_data = requests.get(base_url + request_url, cookies=cookies)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
NameError: name 'requests' is not defined
>>>
```

Refer to the exhibit. A Python code has been written to query a device. The executed code results in the error shown. Which action resolves the problem?

A import json

B requests(ג€GETג€, base_url + request_url, cookies=cookies)

C pip install requests

D import requests ✓ CORRECT

SUGGESTED ANSWER

The suggested answer is D.

The error message 'NameError: name 'requests' is not defined' indicates that the 'requests' module has not been imported into the code. To resolve this issue, the 'requests' module needs to be imported. By including 'import requests' at the beginning of the script, you ensure that Python recognizes and can use the 'requests' library. Therefore, the correct action to resolve the problem is to import the requests module.

Issued to kamjunhao@gmail.com · October 17, 2026

Which REST architectural constraint indicates that no client context should be stored on the server between requests?

A cacheable

B stateless ✓ CORRECT

C uniform interface

D client-server

SUGGESTED ANSWER

The suggested answer is B.

Stateless constraint in REST indicates that no client context should be stored on the server between requests. Each request from a client to the server must contain all the information needed to understand and process the request. This ensures that each request is independent and the server does not retain any client-specific data between requests.

Issued to kamjunhao@gmail.com · October 17, 2026

Which HTTP code group is issued when a request is received successfully, understood, and processed?

A 2xx ✓ CORRECT

B 3xx

C 4xx

D 5xx

SUGGESTED ANSWER

The suggested answer is A.

The HTTP status code group 2xx indicates that a request has been received successfully, understood, and processed. This group includes statuses such as 200 OK, meaning the request was successful and the server returned the requested resource. Other codes in this group provide additional detail such as 201 Created, indicating that the request has resulted in a new resource being created, and 204 No Content, indicating that the request was successful but there is no content to return.

Issued to kamjunhao@gmail.com · October 17, 2026

A developer creates a web application that receives a username and password and uses them to sync the credentials to other services through HTTPS. API keys to services are part of the configuration files of the application, but the credentials to the database that stores the synchronization logs are obtained through an external vault service. What is the security issue in this scenario?

A Communication between the application and the services is not encrypted.

B The database credentials should be stored in the configuration files so that they are secured on the same server.

C The API keys are stored in the configuration files but should be stored in the vault service. ✓ CORRECT

D The synchronization logs should be encrypted and not stored in a relational database.

SUGGESTED ANSWER

The suggested answer is C.

The security issue in this scenario is that the API keys are stored in the configuration files but should be stored in the vault service. Storing API keys in configuration files compromises their security, as these files can be accessed by unauthorized users if the system is compromised. Using a vault service to store sensitive information like API keys is a more secure approach, as it provides better protection and management of these secrets.

Issued to kamjunhao@gmail.com · October 17, 2026

```
#import dnac sdk
from dnacentersdk import DNACenterAPI
#connection to dnac

username = 'user1'
password = 'secret'
base_url = 'https://192.168.1.1'

dnac = DNACenterAPI(username=username, password=password, base_url=base_url,
version='1.3.3', verify=False)

# Get list of ports on device using get all interfaces API call
port_list_1 = [item['portName'] for item in
dnac.devices.get_all_interfaces()['response']
                                if 'da4606c3-63ad-4ed4-8f35-6bfec7c2df04' in
                                item['deviceId']]

# Sort list items
port_list_1.sort()
# Check the number of ports on the device
print(f'The list of interfaces contains {len(port_list_1)} items.')

# Output
# The list of interfaces contains 38 items.


# Get list of interfaces using the get interface info by device id api call
port_list_2 = [item['portName'] for item in
    dnac.devices.get_interface_info_by_id('da4606c3-63ad-4ed4-8f35-
    6bfec7c2df04')['response']]

# Sort list items
port_list_2.sort()
# Check the number of ports on the device
print(f'The list of interfaces contains {len(port_list_2)} items.')

# Output
# The list of interfaces contains 54 items.
```

Refer to the exhibit. An engineer writes a Python script that uses two different API calls to query Cisco DNA Center for the number of interfaces on a device. The engineer notices that the two API calls produce different results. Why are fewer interfaces returned for the same device when the 'get_all_interfaces' API call is used?

A times out

B rate limit

C pagination ✓ CORRECT

D invalid password

SUGGESTED ANSWER

The suggested answer is C.

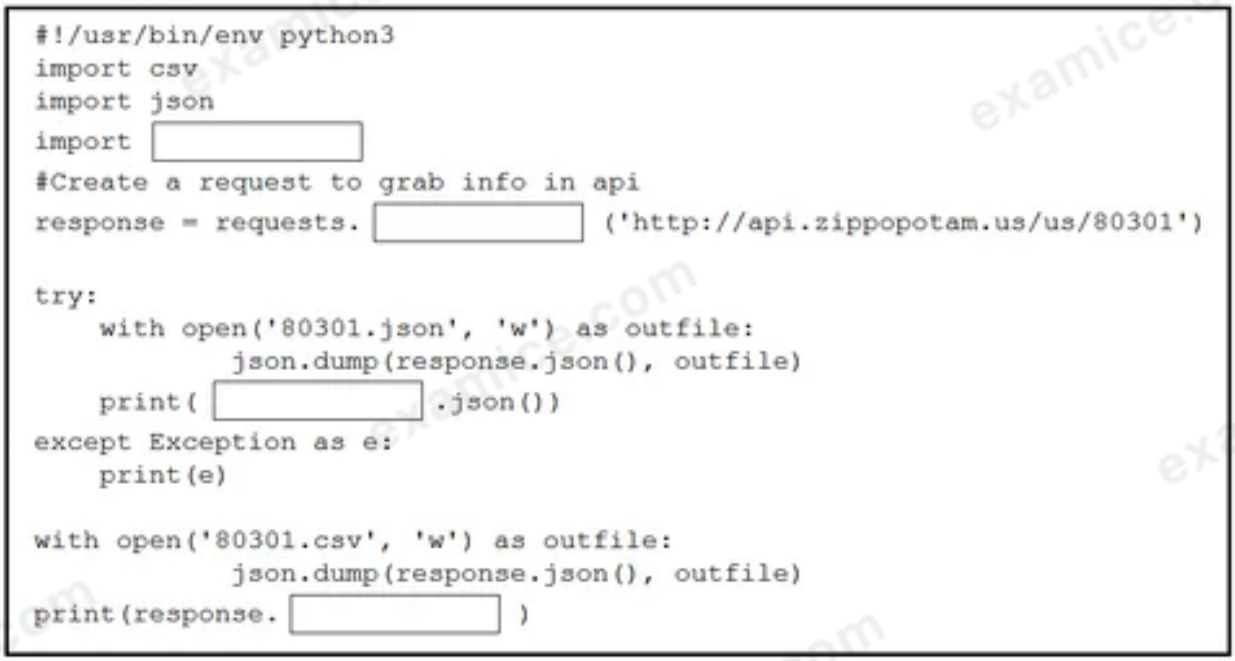
The reason for fewer interfaces being returned when the 'get_all_interfaces' API call is used is likely due to pagination. In API calls, pagination is a technique used to divide the response into multiple pages or chunks, especially when dealing with large datasets. If the script does not handle pagination correctly, it might only fetch the first page of results, resulting in an incomplete list of interfaces. This explains why 'get_all_interfaces' might show fewer items compared to 'get_interface_info_by_id', which may not use pagination or may handle it differently.

Personal copy · kamjunhao@gmail.com

DRAG DROP -

Drag and drop the code snippets from the bottom into the Python script to write API output to a csv file. Not all options are used.

Select and Place:



response	requests	request
get	status_code	status

SUGGESTED ANSWER

```
#!/usr/bin/env python3
import csv
import json
import [requests]
#Create a request to grab info in api
response = requests.[get] ('http://api.zippopotam.us/us/80301')

try:
    with open('80301.json', 'w') as outfile:
            json.dump(response.json(), outfile)
    print([response].json())
except Exception as e:
    print(e)

with open('80301.csv', 'w') as outfile:
            json.dump(response.json(), outfile)
print(response.[status_code])
```

request

status

DRAG DROP -

Drag and drop the code snippets from the bottom to the blanks in the code to test the API response through the Python unittest library. Not all options are used.

Select and Place:

```
import unittest
import json
from apidogs import app

class TestsAPIDogs(unittest.TestCase):

    def setUp(self):
        self.app = app.test client()

    # Check status response of API
    def test status(self):
        resp =self.app.get('/dogs/all')
        status_code = response. [          ]
        self.assertEqual(status_code, [          ] )

    # Check for content type is in application/json
    def test content type(self):
        resp =self.app.get('/dogs/all')
        self.assertEqual(resp. [          ] ,'application/json')

    def test item not exist(self):
        resp =self.app.get('/dogs?id= [          ] ')
        self.assertEqual(response.status code,404)

    if name == '  main  ':
        unittest.main()
```

403	content	status_code
200	none	content_type
	status	

SUGGESTED ANSWER

```
import unittest
import json
from apidogs import app

class TestsAPIDogs(unittest.TestCase):

    def setUp(self):
        self.app = app.test client()

    # Check status response of API
    def test status(self):
        resp =self.app.get('/dogs/all')
        status_code = response. [ none ]
        self.assertEqual(status_code, [ 200 ] )

    # Check for content type is in application/json
    def test content type(self):
        resp =self.app.get('/dogs/all')
        self.assertEqual(resp. [ status_code ] ,'application/json')

    def test item not exist(self):
        resp =self.app.get('/dogs?id= [ 403 ] ')
        self.assertEqual(response.status code,404)

    if name == '  main  ':
        unittest.main()
```

[content]

[content_type]

[status]

DRAG DROP -

```
GET getNetworkHttpServers
https://api.meraki.com/api/v0/networks/:networkId/httpServers
List the HTTP servers for a network

AUTHORIZATION
API Key

This request is using an authorization helper from collection Meraki Dashboard
API
HEADERS
Accept
*/*
```

Refer to the exhibit. A developer is creating a Python script to obtain a list of HTTP servers on a network named office_east by using the Cisco Meraki API. The request has these requirements:

- MUST time out if the response is not received within 2 seconds.
- MUST utilize client certificates and SSL certificate verification.
- MUST utilize basic authentication that uses a username of admin and a password of cisco.
- MUST save the response to an object named response.

Drag and drop the code snippets from the bottom onto the blanks in the code to meet the requirements. Not all options are used.

Select and Place:

```
import requests
from requests.auth import HTTPBasicAuth
network = 'office east'
url = 'https://api.meraki.com/api/v0/networks/{}/httpServers'.format (network)
[          ] = requests.get(url=url, headers={ 'Accept': '*/*'},
verify='/etc/pki/tls/certs/ca.pem',
timeout= [          ] ,
[          ] =('/etc/pki/tls/certs/client.pem',
'/etc/pki/tls/certs/client.key'),
auth= [          ] ('admin', 'cisco'))
response.status_code
```

token	response	2
HTTPTokenAuth	HTTPBasicAuth	cert

SUGGESTED ANSWER

Reference:
https://community.cisco.com/t5/networking-blogs/authenticating-rest-api-calls-to-dnac-getting-started/ba-p/3658058

```
import requests
from requests.auth import HTTPBasicAuth
network = 'office east'
url = 'https://api.meraki.com/api/v0/networks/{}/httpServers'.format (network)
response = requests.get(url=url, headers={ 'Accept': '*/*'),
verify='/etc/pki/tls/certs/ca.pem',
timeout= 2 ,
cert =('/etc/pki/tls/certs/client.pem',
'/etc/pki/tls/certs/client.key'),
auth= HTTPBasicAuth ('admin', 'cisco'))
response.status_code
```

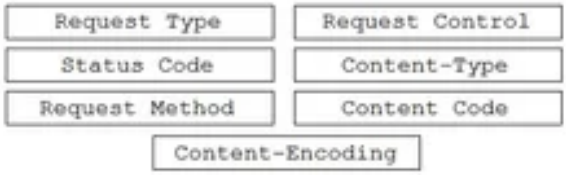
token
HTTPTokenAuth

Personal copy · kamjunhao@gmail.com

DRAG DROP -

Drag and drop the code snippets from the bottom to the blanks in the code to complete the HTTP response. Not all options are used.

Select and Place:



SUGGESTED ANSWER